

Ex Machina: Financial Stability in the Age of AI

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Abstract

We study financial decision making by reinforcement learning algorithms and large language models (LLMs) in a standard mutual fund redemption model with strategic complementarities and multiple equilibria. Both AI architectures depart from the efficient benchmark, and trigger financial fragility, but through distinct mechanisms. Reinforcement learning (Q-learning) algorithms redeem too aggressively due to a systematic learning bias. LLMs struggle to coordinate because they hold different beliefs about how others will act. We characterize the mechanisms behind these failures, showing formally why Q-learning can produce outcomes outside Nash equilibrium and using the reasoning of LLMs to understand why coordination breaks down and how it might be improved. Our results show that the design of AI systems can shape collective outcomes and, ultimately, financial stability.

Keywords: artificial intelligence; financial stability; strategic uncertainty; coordination games; Q-learning; large language models.

JEL classifications: G01, G23, C63, C72.

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I. Introduction

Financial decisions are increasingly being delegated to AI systems. Reinforcement learning algorithms already execute a large share of institutional order flow, with algorithmic trading estimated to account for 60–70 percent of U.S. equity trading volume (Foucault et al., 2025). AI is also rapidly expanding into retail investment (Even-Tov et al., 2025).¹ As adoption grows, AI agents increasingly interact with each another and their decisions become inter-dependent. In this context, when strategic interactions shape financial outcomes, episodes of instability such as fire sales (Brunnermeier and Pedersen, 2009), financial market runs (Bernardo and Welch, 2004), and bank runs (Goldstein and Pauzner, 2005) may emerge as a result of coordination failures. Yet, despite the growing role of AI in trading and investment, we still know little about how AI systems interact in strategic financial settings and what this means for financial stability. Our paper aims to address this gap.

We study two canonical AI decision architectures, namely, reinforcement learning (Q-learning) algorithms, and reasoning-based large language models (LLMs). The setting is a mutual fund redemption model with strategic complementarities and multiple equilibria. Holding the underlying economic environment, payoffs, and incentives fixed, we find that AI architecture shapes aggregate outcomes. Neither architecture consistently selects the efficient equilibrium, leading instead to outcomes characterized by financial fragility. Q-learning investors redeem too much and converge on outcomes that leave them individually worse off. LLM investors, by contrast, fail to coordinate, making individual decisions difficult to predict. The two types of “misbehavior” have different underlying mechanisms. Q-learning is driven by a systematic learning bias, while coordination between LLM investors breaks down because they form different beliefs despite receiving the same prompt.

To our knowledge, this is the first paper to compare different AI decision architectures in a setting with multiple equilibria. Our central result is that architecture shapes equilibrium selection. For Q-learning, we show formally how realized losses, exploration, and counterfactual learning generate excessive redemptions and can push outcomes away from Nash equilibrium. For LLMs, we develop a framework that treats their generated reasoning as data, allowing us to recover the beliefs behind their decisions and identify why coordination breaks down.²

We ground our analysis in the canonical mutual fund redemption model of Chen et al.

¹Recent survey evidence indicates that globally around one in five retail investors use AI tools to select or adjust holdings (eToro, 2025). Moreover, large investment platforms now allow investors to delegate trading directly to autonomous AI agents (Assia, 2026).

²LLMs, and deep learning models more broadly, are often described as “black boxes” because the processes behind their decisions are not well understood. This lack of transparency is considered an important source of risk (Brunnermeier, 2026; Cecchetti et al., 2025).

(2010), which is a parsimonious variant of the models by [Diamond and Dybvig \(1983\)](#) and [Goldstein and Pauzner \(2005\)](#). In the model, investors choose whether to redeem their shares before the fund’s investment project matures. Their payoffs depend on both economic fundamentals and the actions of other investors. Strategic complementarities imply that the incentive to redeem increases when more investors redeem as well. Stronger fundamentals reduce the incentives to redeem. We model the fundamental as the probability with which the fund’s investment yields a positive return and that investors who stay are repaid. It therefore captures default risk in the fund’s portfolio.

Under complete information, the model produces three regions. When the fundamental is weak, it is a strictly dominant action to redeem, and the unique equilibrium is that all redeem. When the fundamental is strong, staying is strictly dominant, and all investors stay. For intermediate fundamentals, both outcomes are equilibria, so investors may either all redeem or all stay. The efficient equilibrium prescribes investors to redeem only when the fundamental is weak, and stay otherwise.

These equilibrium regions provide us with a benchmark for evaluating AI behavior. We replace investors with Q-learning algorithms and LLMs and run simulation-based experiments across different fundamentals. We define “misbehavior” as a departure from the efficient equilibrium that increases financial fragility. We first study Q-learning and then LLM investors, focusing on why each one departs from the benchmark.

Q-learning’s redemption bias is the result of two effects. The first is a *hot stove effect*: the accumulation of losses following the materialization of default risk, combined with limited algorithmic exploration, biases the learning dynamics and pushes Q-learning investors toward inefficient redemptions.³ We formally prove that this bias is unavoidable in the presence of limited exploration, stochastic payoffs and constant learning gains. The result is not limited to settings with multiple equilibria. It also holds when there is a unique dominant strategy Nash equilibrium.

The second channel through which the Q-learning bias operates is an *exploration effect*, which we expose in an exercise removing default risk. Using expected rather than realized payoffs reduces the hot stove effect, but the algorithm still fails to reach the efficient outcome. Instead, Q-learning converges to a strategy that best-responds to other investors randomizing between redeeming and staying. This outcome is consistent with that obtained under level-1 thinking, which is a model of bounded rationality ([Nagel, 1995](#)). We formally prove that this behavior emerges when the exploration phase is sufficiently long relative to how quickly learning converges. Through these two effects, Q-learning investors coordinate,

³The expression “*hot stove effect*” was coined by [Denrell and March \(2001\)](#) to refer to situations where agents over-react to negative experiences by entirely avoiding similar future situations.

but on a privately and socially suboptimal action. This stands in contrast to the algorithmic collusion literature, where algorithms coordinate on privately optimal actions and earn supra-competitive profits (Calvano et al., 2020; Colliard et al., 2025; Dou et al., 2025a). Here, coordination instead leads them to a harmful outcome.

LLM investors, by contrast, reason from the information provided in the prompt rather than update their behavior through realized payoffs. They are therefore not subject to the same hot stove effect.⁴ However, the theoretical indeterminacy that gives rise to multiple equilibria in the intermediate region of fundamentals manifests itself as belief heterogeneity: identically prompted LLMs form different beliefs about how other investors will act. In this region, the same information can support several internally consistent beliefs about how other investors will behave. This variation in beliefs makes it difficult for LLM investors to coordinate. Importantly, this result is inconsistent with mere retrieval of known solutions from training data, suggesting that LLM investors are engaging in genuine strategic reasoning rather than behaving as “stochastic parrots” (Bender et al., 2021).

Having shown that the two architectures misbehave for different reasons, we next consider targeted changes to improve their behavior. This matters because safeguards that work for one type of AI may fail for another, or may replace one problem with a different one. For Q-learning, we introduce synchronous updating, which allows the algorithm to update its estimate of both actions after each round. We prove that this eliminates the hot stove effect. However, it also makes the estimated value of staying highly volatile, preventing investors from settling on an equilibrium. This result complements Asker et al. (2022) and Banchio and Mantegazza (2025), who show that synchronous learning eliminates collusive and socially inefficient behavior in games with a unique Nash equilibrium. In our setting with multiple equilibria, it removes the learning bias but creates a new coordination problem.

For LLMs, we test two interventions designed to reduce differences in beliefs. The first introduces uncertainty about the fundamental. Each LLM receives a noisy private signal, giving agents a common basis for forming expectations while preserving some disagreement. The introduction of noise induces convergence to a common threshold strategy, and leads to aggregate redemptions that closely match the corresponding global game benchmark (Carlsson and Van Damme, 1993; Morris and Shin, 1998). Imperfect private signals, thus, emerge as a focal point for LLMs to choose a common strategy profile, which improves coordination and stability. Second, we allow the LLMs to observe previous outcomes in a repeated game set up. In this case, we show that history serves as a focal point and so coordination emerges (Argenziano and Gilboa, 2012). Whether LLMs coordinate on the good (no redemption)

⁴LLMs generate text by predicting the next token based on the context provided and patterns learned during training.

or bad (universal redemption) equilibrium depends on the exact history: observing fewer redemptions leads to coordination on the efficient equilibrium, while the opposite to the socially and privately suboptimal one.

Taken together, our results show that financial stability depends not only on economic incentives but also on the type of AI system making the decisions. Learning-based systems can coordinate on inefficient outcomes, while reasoning-based systems can fail to coordinate when no clear focal point exists. AI therefore does not simply implement standard equilibrium predictions; it shapes which equilibrium emerges and, in turn, how financial fragility develops.

Outline. The rest of the paper proceeds as follows. Section II summarizes the related literature. In Section III, we present a stylized theoretical framework of mutual fund redemptions. In Section IV, we describe how we implement the two AI architectures. The baseline results highlighting the misbehaviors are reported in Section V. Remedies are analyzed in Section VI. Section VII concludes.

II. Related Literature

Our paper contributes to a growing body of work on the economics of AI, and in particular on strategic behavior in algorithmic systems. A central finding in this literature is that algorithms can learn to sustain tacit collusion and earn supracompetitive profits without communicating explicitly. This result was first shown by [Calvano et al. \(2020\)](#) in simulation experiments with Q-learning algorithms in a repeated pricing oligopoly. Since then, it has been documented in other settings, e.g., auctions ([Banchio and Skrzypacz, 2022](#)), and shown to be rooted in the design of the algorithm (in particular, the learning protocol) ([Asker et al., 2022, 2024](#)). Algorithmic collusion and the mechanisms behind it have also been explored in financial market applications, where asymmetric information, noise, and stochastic payoffs add further complexity. Collusion has been shown to emerge among algorithmic market makers ([Colliard et al., 2025; Guarino et al., 2025](#)), and AI-powered informed speculators ([Dou et al., 2025a](#)). More recently, [Dou et al. \(2025b\)](#) extend this literature by showing that algorithms can also learn trading strategies that manipulate prices.

Our analysis differs from this literature in two important ways. First, we study both Q-learning algorithms and LLMs, which represent two distinct approaches to AI decision-making. Second, we examine a static coordination game with strategic complementarities and multiple equilibria, placing the problem of equilibrium selection at the center of the analysis. Our contribution to the AI collusion literature is twofold. First, we show that algorithms do not always select the payoff-dominant equilibrium. In our setting, coordination can instead

lead AI investors to a stable but privately inefficient outcome, in contrast to the tacit collusion results in [Calvano et al. \(2020\)](#). Second, we complement simulation results with formal proofs of the mechanisms driving Q-learning algorithms to coordinate on a specific equilibrium. In line with recent analyses ([Banchio and Mantegazza, 2025](#); [Guarino et al., 2025](#); [Dou et al., 2025c](#)), which provide an analytical characterization of the algorithms’ behavior and the equilibria they converge to, our paper methodologically advances the understanding of how Q-learning algorithms behave and how their learning is influenced by specific characteristics of the economic environment where they operate.

[Horton et al. \(2023\)](#) introduced the idea of using LLMs as simulated economic agents, or *homo silicus*, whose information, preferences, endowments, and incentives can be varied across economic settings. Our framework builds on this approach by recovering the beliefs and reasoning behind LLM decisions, not just their observed choices. Similarly, [Cook et al. \(2026\)](#) use revealed preference methods and structural estimation to identify LLM preferences in allocation and job search tasks, showing that these preferences are systematic but sensitive to framing and model interventions.⁵ Building on the broader idea of using LLMs as economic agents, a growing literature simulates the behavior of investors ([Fedyk et al., 2024](#)), bank depositors ([Kazinnik, 2023](#)), and loan officers ([Cook and Kazinnik, 2025](#)). LLMs can also serve as proxies for survey respondents ([Hansen et al., 2024](#)) and reproduce human-like macroeconomic expectations ([Bybee, 2023](#); [Zarifhonarvar, 2024](#)). Within finance, [Lopez-Lira \(2025\)](#) introduce an open source framework in which LLM trading agents interact with value investors, momentum traders, market makers, and contrarians. [Gao et al. \(2024\)](#) show that heterogeneous LLM investors generate price dynamics resembling those observed in financial markets. [Bhagwat et al. \(2025\)](#) elicit beliefs from LLM investor personas and find that disagreement about firm news tracks abnormal trading volume and predicts subsequent returns.

We contribute to this literature in two ways. First, we compare LLM and Q-learning investors in a canonical coordination game, clarifying how explicit reasoning, rather than learning, shapes equilibrium selection. Second, we introduce a framework for analyzing LLMs’ latent reasoning. By treating chain-of-thought text as data, we recover beliefs about others’ actions, the justifications for those beliefs, and the links from justifications to beliefs and decisions. Because we infer beliefs from the reasoning that produces decisions, rather than

⁵Our approach here differs from agent-based modeling (ABM), which has been used in economics and finance to study the emergent behavior of heterogeneous, boundedly rational interacting agents that cannot be captured using a representative agent framework ([Farmer and Foley, 2009](#); [Battiston et al., 2012](#)). Whereas ABMs specify behavioral rules for interacting agents to study aggregate dynamics, the approach here treats the LLM itself as an object of study and seeks to understand how AI agents form beliefs, reason, and make decisions.

through direct elicitation, our approach is less vulnerable to post-hoc rationalization and helps explain why LLM investors behave as they do.

Our analysis shows that belief heterogeneity prevents LLMs’ coordination when multiple equilibria exist. Without a shared focal point, identically prompted LLMs form different but internally consistent beliefs about others’ behavior. With noisy private signals, however, they converge on a common threshold strategy. This parallels [Heinemann et al. \(2004\)](#), who find that private information moves human behavior closer to the global-games solution, providing a useful benchmark for interpreting LLM responses to strategic uncertainty. Together with our reasoning analysis, these results suggest that LLMs can apply sophisticated game theoretic logic when the information structure guides belief formation.

Within the nascent literature on LLMs as decision-making agents, our paper is closest to [Fish et al. \(2024\)](#). Our results on LLMs complement their analysis, which shows that LLM-based pricing agents learn to collude in a repeated Bertrand competition game. We show that learning is not the only path to coordination. With fundamental uncertainty, i.e., when LLM investors receive private signals, they coordinate even in a one-shot setting. This suggests that the information environment itself can serve as a coordination device for reasoning-based AI. Furthermore, collusion (i.e., coordination on the privately optimal outcome) is not automatically achieved when extending the environment into a dynamic setting. Equilibrium selection is obtained by allowing LLMs to observe past history, but history itself determines which equilibrium LLMs coordinate on.

Our paper also contributes to the emerging literature on AI and financial stability. This topic has attracted much attention from policymakers ([Shabsigh and Boukherouaa, 2023](#); [Aldasoro et al., 2024](#); [Financial Stability Board, 2024](#); [Leitner et al., 2024](#); [Bank of England, 2025](#)), discussing several channels through which AI may destabilize the financial system, including procyclicality, unknown unknowns, herding, increased exposure to cyber risks. [Danielsson and Uthemann \(2025\)](#) formalize some of these ideas using a global games framework, showing that AI’s superior information processing, reliance on common data sources, and speed advantages can make transitions between stable and crisis states more abrupt. We advance this literature by studying AI behavior in a canonical coordination game grounded in economic theory. This setting allows us to isolate the forces shaping Q-learning and LLM decisions and to show how they affect equilibrium selection and financial fragility. In this respect, our study is also related to [Yang \(2024\)](#), who shows that Q-learning can trigger currency attacks in a two-agent speculative attack setting. We instead study a mutual fund redemption game with many interacting investors. We formally characterize Q-learning behavior and compare it directly with LLM behavior, allowing us to show how AI architecture and the economic environment jointly shape financial fragility.

Finally, we also contribute to the literature on financial fragility. Following the seminal work of [Diamond and Dybvig \(1983\)](#), this literature has extended to other financial institutions subject to strategic uncertainty, such as mutual funds ([Chen et al., 2010](#)), hedge funds ([Liu and Mello, 2011](#)), credit markets ([Bebchuk and Goldstein, 2011](#)), life insurers ([Foley-Fisher et al., 2020](#)) and stablecoin issuers ([Gorton et al., 2025](#)), among others. Despite rich theory, credible empirical evidence remains scarce, with [Goldstein et al. \(2017\)](#) and [Chen et al. \(2024\)](#) as notable exceptions. Our contribution here is that we use AI agents to test theories of financial fragility in a setting where information, payoffs, and strategic interactions can be precisely controlled. This allows us to isolate how economic fundamentals and AI design shape financial risk.

III. Theoretical Benchmark

We build on [Chen et al. \(2010\)](#) and characterize the equilibria in a stylized mutual fund redemption game. The economy extends over two dates, $t = 1$ and $t = 2$, and consists of a mutual fund and an integer number, $N > 2$, of risk-neutral investors. Each investor holds one share in the mutual fund. Thus, the initial size of the fund is N .

At $t = 1$, the book value of the fund's investments is $R_1 N$, which is common knowledge, with $R_1 \geq 1$. At $t = 2$, the per-unit investment return is instead:

$$R_2(\theta) = \begin{cases} R & \text{with prob. } \theta, \\ 0 & \text{with prob. } 1 - \theta, \end{cases} \quad (1)$$

where $\theta \in [0, 1]$ is a uniformly distributed random variable drawn at the start of $t = 1$ and represents the strength of the economic fundamentals. Stronger fundamentals (higher θ) imply a higher probability that the fund's investments yield a high return and hence lower default risk for investors.⁶

At the beginning of $t = 1$, a number A of active investors observe θ and choose whether to redeem their shares or hold them until $t = 2$. Investors who redeem receive the current value R_1 . However, in order to service redemptions, the fund must liquidate its investments, which is costly. Specifically, to raise R_1 , the fund must liquidate $(1 + \lambda)R_1$ worth of assets, where $\lambda > 0$ is a measure of illiquidity. The higher is λ , the more assets must be liquidated to service redemptions. Following [Chen et al. \(2010\)](#), we assume that $A \leq \frac{N}{1+\lambda}$, implying that the mutual fund has enough resources to meet all redemptions.

⁶In [Chen et al. \(2010\)](#), the variable θ is a measure of the performance of the mutual fund. For the purpose of our exercise, the precise definition is immaterial.

After servicing redemptions at $t = 1$, the fund re-invests its remaining resources. Denoting the number of investors who redeem at $t = 1$ by $W \leq A$, an investor who decides to stay invested expects to receive a pro-rata share of the fund's return at $t = 2$, which is given by:

$$R_2(\theta) R_1 \frac{N - W(1 + \lambda)}{N - W}. \quad (2)$$

Thus, an investor's payoff from staying is increasing in the strength of economic fundamentals, θ , and decreasing in both asset illiquidity, λ , and the number of redemptions, W . It is useful to define the critical mass of redemptions $\widehat{W}(\theta)$ at which an investor is just indifferent in expectations between staying and redeeming, as the solution to:

$$1 = R\theta \frac{N - \widehat{W}(1 + \lambda)}{N - \widehat{W}}. \quad (3)$$

The threshold $\widehat{W}(\theta)$ is increasing in θ , as stronger fundamentals increase the value of staying and therefore requires a higher critical mass of redemptions to make investors indifferent between staying and redeeming.

We focus on characterizing equilibria in pure strategies. Following standard lines of reasoning (Morris and Shin, 1998), we can partition the support for the fundamental into three intervals, which we report in Lemma 1 and Figure 1.

Lemma 1: *The equilibrium outcomes depend on the realization of θ . For sufficiently extreme values of θ , there exist a unique Nash equilibrium in pure strategies, while for intermediate values multiple equilibria arise. Specifically,*

- if $\theta < \underline{\theta} = \frac{1}{R}$, the unique equilibrium is that all investors redeem at $t = 1$;
- if $\theta > \bar{\theta} = \frac{1}{R} \frac{N - (A - 1)}{N - (A - 1)(1 + \lambda)}$, the unique equilibrium is that all investors stay until $t = 2$;
- if $\theta \in [\underline{\theta}, \bar{\theta}]$, both all-redeem and all-stay are equilibria.

An important prerequisite for multiple equilibria is costly liquidation. When $\lambda > 0$, the expected payoff difference between staying until $t = 2$ and redeeming at $t = 1$ decreases monotonically with the number of investors who redeem. Hence, investors' decisions are strategic complements, i.e., the incentive to redeem is increasing in the belief that others will redeem as well. Crucially, for $\theta \in [\underline{\theta}, \bar{\theta}]$ the belief that all others will redeem and all others will stay are both rationalizable leading to multiple self-fulfilling equilibria.

Consistent with the literature on endogenous withdrawal decisions (e.g., Goldstein and Pauzner, 2005), we take the investors' decisions as a constraint in the planner's optimization.



Figure 1: Equilibria without fundamental uncertainty.

Proposition 1 describes the benchmark against which we compare algorithmic behavior in the incoming sections.

Proposition 1: *The constrained efficient equilibrium is for all investors to redeem at $t = 1$ if $\theta < \underline{\theta}$ and stay otherwise.*

IV. Experimental Design

We replace the investors of the theoretical model with algorithmic counterparts. We consider two distinct types of AI-agents: Q-learning investors, who optimize through trial-and-error reward updating, and LLM investors, who reason contextually via chain-of-thought inference. The two classes differ not only in how they behave but also in how we can characterize that behavior. For Q-learning investors, the learning dynamics are analytically tractable: we establish their limiting behavior in closed form and use simulation to confirm the result and probe their robustness. For LLM investors, no such characterization is available, and simulation is the primary evidence. Both classes are evaluated under a common economic environment. There are $A = 30$ active investors out of a total of $N = 50$. The parameters governing investment returns are $R_1 = 1$ and $R = 2$. For the liquidation cost, we set $\lambda = 0.25$. The fundamental is drawn independently each time the game is played from an equally spaced twenty-five-point grid, Θ , spanning the interval $[0.4, 1]$ that covers both the efficient redemption bound $\underline{\theta} = 0.5$ and the no redemption bound $\bar{\theta} = 0.77$.

A. Q-learning

Q-learning is a transparent and economically interpretable reinforcement learning algorithm. With no prior knowledge of the payoff structures, agents learn the values of choosing different actions through trial-and-error interaction with the environment and by receiving rewards and penalties. An agent's optimal action is the one that maximizes the reward across independent, one-shot interactions.

In our set-up, the algorithm has four components:

1. **States:** A state is payoff relevant information observed by each algorithm before choosing its action. In our setting, the only such information is the fundamental, so with a slight abuse of notation, we define θ to be the state drawn from the grid Θ .
2. **Actions:** The action set is binary, $\mathcal{A} = \{a_R, a_S\}$, where a_R denotes “redeem” and a_S denotes “stay”.
3. **Rewards:** The reward function $\pi(\theta, a)$ assigns a payoff based on the realization of the fundamental θ and the chosen action a .⁷ Crucially, Q-learning investors do not observe the underlying mapping $(\theta, a) \mapsto \pi$, but only the realized rewards.
4. **Episodes:** Learning unfolds over T episodes. In each episode, the fundamental is drawn i.i.d, so each state is realized with probability $\nu \equiv 1/|\Theta| > 0$. Q-learning investors observe their state, choose an action, and update their Q-values based on the realized payoff.

Each Q-learning investor $i = 1, \dots, A$ starts episode $t = 1, \dots, T$ with a Q-matrix $Q_{i,t} \in \mathbb{R}^{|\Theta| \times 2}$, with rows for states and columns for actions. The fundamental is drawn i.i.d. and the state, θ is determined. Investors choose their actions according to an ε -greedy policy: with probability $\varepsilon_t = \beta^t$, the Q-learning investor explores by randomizing between a_R and a_S , while with the complementary probability $1 - \varepsilon_t$, the Q-learning investor exploits by choosing the action that maximizes $Q_{i,t}(\theta, a)$.

If the chosen action is $a_{i,t}^* = a_R$, the reward is $\pi(\theta, a_R) = R_1$. If instead $a_{i,t}^* = a_S$, the reward depends on the true fundamental and the actions of the other Q-learning investors, and is given by Equation (2). The Q-matrix is then updated as

$$Q_{i,t+1}(\theta, a_{i,t}^*) = (1 - \alpha)Q_{i,t}(\theta, a_{i,t}^*) + \alpha \pi(\theta, a_{i,t}^*), \quad (4)$$

where $\alpha \in [0, 1]$ is the learning rate: higher α places more weight on new rewards, while lower α smooths learning over past experience.⁸ Figure 2 summarizes the iterative process.

In our simulations, we set $\beta = 0.99999$ and $\alpha = 0.1$. We run 25 independent rounds of training, each with $T = 500,000$ episodes. During each training round, the initial Q-values are drawn at random from the unit interval. For these hyperparameters, each Q-learning

⁷Note that $\pi(\theta, a)$ implicitly depends on the actions of other investors through Equation (2), but we suppress this dependence in the notation since Q-learning treats other agents’ actions as part of the environment.

⁸Since the state is drawn i.i.d in each episode and investors’ actions today do not impact the state that materializes tomorrow, excluding the forward-looking component that often appears in Q-learning update rules is without loss of generality. Each investor therefore solves a bandit problem for each value of θ (Langford and Zhang, 2008).

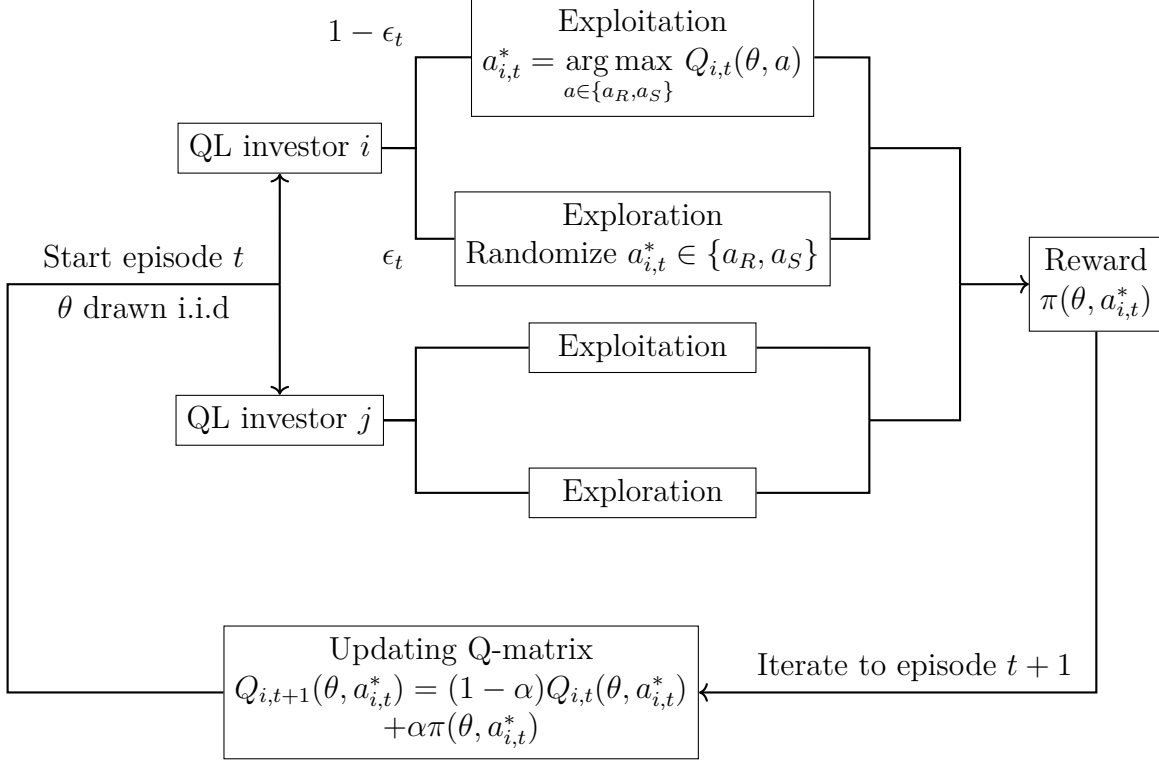


Figure 2: Iterative process of Q-learning.

investor experiments approximately 100,000 times. Assuming a decaying exploration rate is standard in the literature (e.g., Colliard et al., 2025) and reflects the idea that exploration corresponds to training, which is costly and therefore limited.

B. Large Language Models

LLMs are trained to predict the next token in a sequence. Drawing on the vast corpora of text used for training, these models attempt to determine the meaning and context of prompts, or input text, given to them in order to improve their predictions and responses. In addition, some models, like OpenAI’s o3, Google’s Gemini 2.5 Pro, and DeepSeek’s R1 can “think out loud”, i.e., produce a sequence of latent reasoning steps linking the prompts given to them, to assumptions, intermediate steps of calculations to the final output. As such, LLMs are capable of instantaneously adapting their behavior when the information they are presented with changes. In our experiments, we use DeepSeek’s R1-0528 to model the LLM investors. A key advantage of using this LLM is that the full chain-of-thought reasoning can be extracted and analyzed, unlike with other reasoning-based LLMs. DeepSeek’s models are also open weight, which supports transparency and replicability.⁹

⁹For a broader discussion on the use and evaluation of open weight LLMs, see Cook et al. (2023).

The experiment proceeds in two stages. First, we construct the prompts. Each prompt gives the investor a concise description of the economic environment, including the objective, payoff structure, and available information.

Second, the prompt is instantiated with the relevant parameters and submitted to the LLM via API calls. Every investor is represented by a separate, independent API call. To encourage consistent behavior, we set the temperature hyperparameter to zero, reducing randomness in the outputs.¹⁰ From each call, we record both the latent chain-of-thought reasoning and the final decision. We run three independent rounds of inference to account for any residual randomness and construct confidence intervals around the outcomes.

Finally, to explain the decision-making of LLM investors, we treat the chain-of-thought reasoning text as data, which we then analyze using a LLM (we also use DeepSeek R1-0528) to extract beliefs and reasoning. Our approach proceeds in two steps. First, we prompt the LLM to extract from each investor’s reasoning text: (i) the belief they held about the behavior of others, and (ii) the justification underlying that belief. Second, we ask the same LLM to cluster these into canonical game-theoretic categories. The resulting graph maps justifications to beliefs to decisions, revealing the logical structure of AI-investor reasoning.¹¹ To ensure consistency in the labels for the justifications and beliefs, we jointly cluster the DAGs for all investors across all values of the fundamental.

¹⁰The temperature setting controls the degree of randomness in the model’s responses: a value near zero yields deterministic and focused outputs, while higher values introduce more variation.

¹¹The prompts used to extract beliefs and reasoning are shown in Appendix B.

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1 You are one of A=%d active investors in a mutual fund out of a
   total of N=%d investors. Each investor holds one share. Your
   goal is to maximize your return.
2 If you redeem your share, then you earn R1.
3 If you do not redeem your share (i.e., you stay), then with
   probability  $\theta$  you earn fraction * R1 * R, and otherwise you get 0,
   where:
4 - fraction = (N - W * (1 +  $\lambda$ )) / (N - W) is the fraction of assets
   remaining in the fund after serving redemptions;
5 -  $W \leq A-1$  is the number of other active investors who redeem (the
   remaining N - A investors are passive and never redeem);
6 -  $\lambda$  (lambda) = %f is the illiquidity parameter;
7 - R1 = %f is the value of the share if you redeem;
8 - R = %f is the return earned by the fund from managing its
   portfolio;
9 -  $\theta$  (theta) = %f is the fundamental, which measures the fund's
   performance.
10 Do you choose to redeem your share or stay? State your decision
    with exactly one word: ``redeem'' or ``stay'' using the XML tag <
    decision>...</decision>.

```

Prompt 1: Baseline prompt.

V. Baseline results: Algorithmic misbehaviors

Under the efficient equilibrium, all investors should redeem if and only if the fundamental is below the lower dominance bound, i.e., $\theta < \underline{\theta}$. Building on this, we classify an algorithm as misbehaving if the realized outcome deviates from the frictionless efficient equilibrium. Those are the instances in which individual behavior collectively triggers financial fragility.

Figure 3 plots the share of Q-learning and LLM investors who redeem, as a function of the economic fundamental, θ . Both algorithms adhere to the efficient benchmark when the fundamental is weak, i.e., all Q-learning and LLM investors redeem when $\theta < \underline{\theta}$. But above this threshold, they misbehave and their behaviors differ significantly. Q-learning investors over-redeem: they coordinate on redemption even in ranges of fundamentals where, according to the theory, no redemptions should be observed, i.e., when $\theta > \bar{\theta}$. In contrast, when the fundamental is strong, $\theta > \bar{\theta}$, all LLM investors stay and adhere to the efficient benchmark.

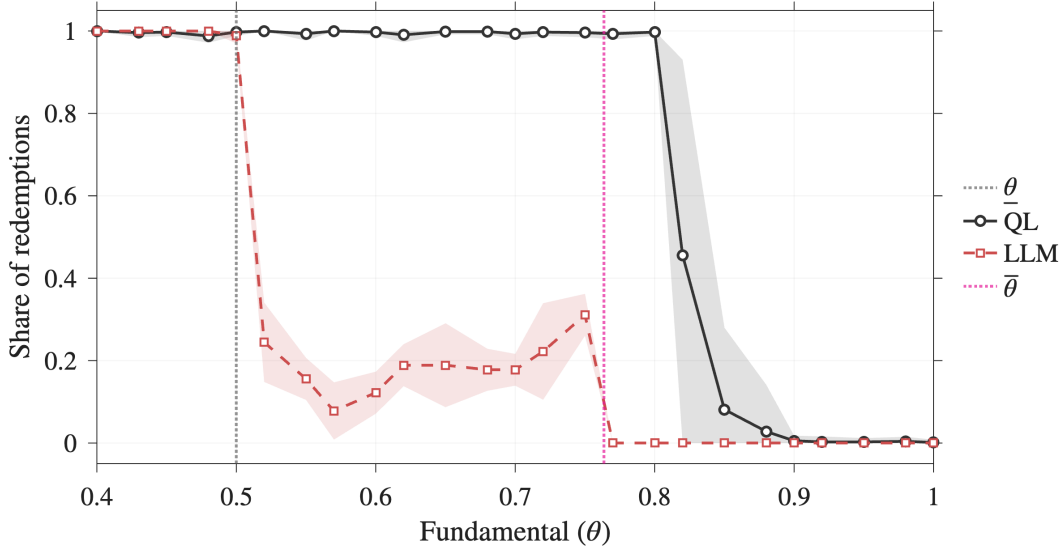


Figure 3: Share of redemptions as a function of the fundamental θ . The line labelled “QL” refers to the outcome with Q-learning investors, while the line labeled “LLM” is with the LLM investors. Shaded areas show the standard deviations constructed across independent runs: twenty-five training rounds for Q-learning investors and three inference rounds for LLM investors. The vertical lines $\underline{\theta} = 0.5$ and $\bar{\theta} = 0.764$ are the bounds defined in Lemma 1.

However, in the intermediate region, they do not coordinate on the all-stay Nash equilibria. Instead, for a given θ , some LLM investors stay while others redeem.

In sum, Q-learning and LLMs exhibit different misbehaviors that lead them to deviate from the efficient equilibrium benchmark. Next, we explain the mechanisms behind these results. For ease of exposition, we start with exploring over-redemption with Q-learning investors and then subsequently turn to the LLM investors’ inability to coordinate.

A. Q-learning: Learning biases

A.1. Hot stove effect

The Q-learning redemption bias stems from how the underlying learning dynamics operate. Consider the case where θ is just above $\bar{\theta}$. Since the fundamental is strong, the probability that the fund’s investments default is low. However, each time a Q-learning investor chooses to stay and default risk materializes, they obtain a zero reward, which revises down the Q-value of staying. Since the reward from redeeming is fixed and certain, following a sufficiently long sequence of consecutive zero-rewards, the investors switch to favoring redeeming. Once an investor switches, they stop updating the Q-value for staying. As a consequence, they incorrectly conclude that the expected reward from staying is below that of redeeming, and

this error is never corrected. The *hot stove effect* has previously been identified to explain other algorithmic biases (Denrell and March, 2001).¹²

As the description above suggests, the hot stove effect requires both a sufficiently long sequence of consecutive default events and that Q-learning investors do not continue to update the Q-values for stay once they switch. While the second property is a consequence of the standard assumption of asynchronous updating (i.e., only the Q-value of the action taken is updated) the first one arises from vanishing exploration. Proposition 2 shows that the hot stove effect is highly pervasive and biases Q-learning investors towards redeeming for all $\theta < 1$.

Proposition 2: Q-learning investors almost surely reach a date in finite time after which they redeem forever for any $\theta < 1$.

In the special case $\theta = 1$, the fund’s investments never default, so staying yields a strictly positive payoff with certainty and emerges as the outcome in line with the efficient equilibrium benchmark. However, for every other value of θ , the hot stove effect dominates and Q-learning investors lock in on redeeming, even where staying is the unique dominant-strategy equilibrium. The mechanism is therefore not one of equilibrium selection: strategic uncertainty and equilibrium multiplicity play little role. This is a learning bias: correct payoff signals are mishandled under updating.

Our findings suggest that Q-learning algorithms may not play the static Nash equilibria. In other papers in the literature (Colliard et al., 2025; Calvano et al., 2020; Dou et al., 2024), this result is interpreted as evidence of Q-learning algorithms tacitly coordinating on outcomes that are socially suboptimal but that deliver higher private payoffs. In our framework instead, Q-learning investors choose actions that yield the lowest (and sure) payoff. The ensuing equilibrium is both socially and privately suboptimal.

A.2. Exploration effect

To further understand the mechanism behind the over-redemption by Q-learning investors, we remove default risk. In this case, we assume that with probability one the payoff from staying is equal to its expected value, i.e., $R_2(\theta) = R\theta$. Importantly, this does not change the efficient benchmark, since risk-neutral investors base their decisions on the expected payoffs. Figure 4 plots the share of redemptions as a function of the fundamental for Q-learning investors. The black line is our baseline result, while the green line represents the case where the reward for staying is the expected payoff. We readily note that in the case without

¹²The name of the effect is in deference to Mark Twain who once wrote, “*Like a cat sitting on a hot stove lid, only to get burned, and learning to never sit on a stove lid again, even a cold one*”.

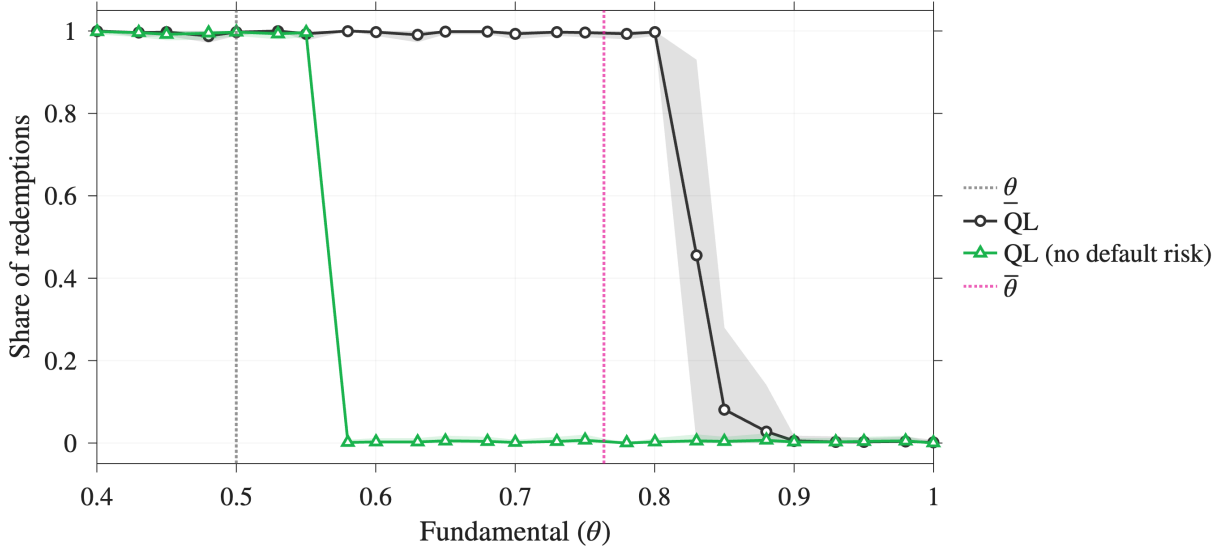


Figure 4: Share of redemptions as a function of the fundamental θ . The line labelled “QL” refers to the baseline case (with default risk), while the line labeled “QL (no default risk)” refers to the exercise where we replace the realized rewards with the expected rewards. Shaded areas show the standard deviations constructed across twenty-five independent training rounds. The vertical lines $\underline{\theta} = 0.5$ and $\tilde{\theta} = 0.764$ are the bounds defined in Lemma 1.

default risk there is a sharp switch in the share of redemptions around $\tilde{\theta} = 0.558$: for $\theta < \tilde{\theta}$, all Q-learning investors redeem, while for larger values of θ , they all stay.

The threshold $\tilde{\theta}$ aligns with the solution to the indifference condition between redeeming and staying, where the expected payoff for staying is calculated assuming that all other investors randomize between redeem and stay with equal probability. This resembles what one obtains using a level-1 reasoning model of bounded rationality (Nagel, 1995). Formally,

$$\tilde{\theta} = \frac{R_1}{\left(\frac{1}{2}\right)^{A-1} \sum_{W=0}^{A-1} \binom{A-1}{W} \left(\frac{N-W(1+\lambda)}{N-W}\right) R_1 R}. \quad (5)$$

Notice that $\tilde{\theta}$ lies strictly above the efficient redemption bound $\underline{\theta} = 0.5$. So, while removing default risk eliminates the hot stove effect, it exposes a residual and more subtle mechanism, which we call the *exploration effect*: the algorithms learn to best-respond to the random play generated by their exploration.

During exploration, every investor randomizes with equal probability, so the environment each Q-learning investor faces is the equal-probability mixing distribution underlying $\tilde{\theta}$. Since the Q-values update at rate α , the investors learn the expected payoffs of this environment provided two conditions hold. First, the influence of the initial Q-matrix must dissipate,

which the constant learning rate guarantees. Second, learning must converge faster than exploration vanishes. If the exploration rate ε_t decays too quickly, investors lock in on half-learned values shaped by initial conditions rather than on the best response to randomizing opponents. When learning is fast relative to the decay of exploration, there is a window in which exploration remains high and the Q-values have already settled onto the equal-randomization best response. Proposition 3 formalizes this result.

Proposition 3: *Suppose that the learning rate α , the exploration rate $\varepsilon_t = \beta^t$, and the uniform probability ν for selecting a state, satisfy:*

$$-\log \beta \ll \alpha/\nu. \quad (6)$$

Then, absent default risk, under the mean-field approximation that becomes exact in the limit $A \rightarrow \infty$, Q-learning investors converge to the threshold policy that redeems for $\theta < \tilde{\theta}$ and stays for $\theta > \tilde{\theta}$, where $\tilde{\theta}$ is given by (5).

Proposition 3 isolates the residual exploration effect under a mean-field approximation where we replace the random, realized Q-values in the update rule with their expected values. The result proves that Q-learning investors converge to an outcome governed entirely by the exploration process, which we characterize in closed form. The result explains the findings in Abada and Lambin (2023) who show via simulations that replacing uniform exploration with “rational” exploration on the Nash action eliminates the non-Nash outcomes.

B. LLM investors: Belief heterogeneity

Figure 3 shows that LLM investors fail to coordinate when the economic fundamentals fall in the range $(\underline{\theta}, \bar{\theta})$, i.e., the region where theory predicts multiple equilibria. By analyzing the chain-of-thought produced by each individual LLM investor, we trace this behavior to *belief heterogeneity*.

Figure 5 presents an illustrative mapping from justifications to beliefs and redemption decisions for $\theta = 0.75$, obtained by analyzing the chain-of-thought reasoning text that each LLM investor produces in a round. Investors exhibit substantial heterogeneity in both justifications and beliefs. Two justification categories were identified: (i) *payoff comparison under uncertainty*, where investors compare payoffs across equilibria under particular beliefs, and (ii) *equilibrium analysis*, where investors reason using formal equilibrium concepts. These justification categories map into three belief categories: (i) *focal point (efficient)*, expecting others to coordinate on the Pareto-optimal equilibrium; (ii) *focal point (safe)*, expecting others to coordinate on the safe outcome; (iii) *probabilistic beliefs*, assuming others redeem or

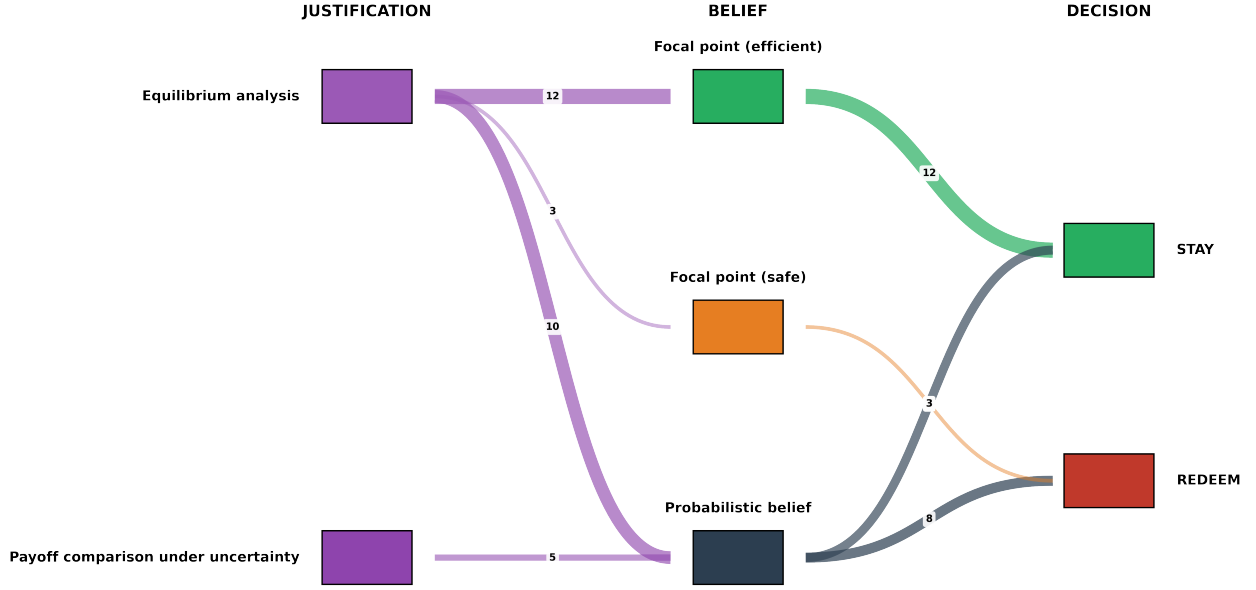


Figure 5: Mapping from justifications to beliefs and redemption decisions, for LLM investors at $\theta = 0.75$. Link widths and numbers indicate investor counts.

stay with specific probabilities. The mapping from beliefs to redemption decisions suggests that the investors correctly play best-responses, i.e., when expecting others to coordinate on the efficient equilibrium they choose to stay, while when they expect others to coordinate on the safe equilibrium, they choose to redeem.

The heterogeneity arises even though all investors receive identical prompts. We set temperature to zero to minimize randomness, but independent queries still produce small differences in the outputs. For intermediate fundamentals, the prompt does not imply a unique way to reason about beliefs and multiple equilibria. Several internally consistent interpretations are possible, and different investors can settle on different ones. As a result, identical prompts can generate heterogeneous beliefs even when each investor’s reasoning is internally consistent.

Figure 6 plots the evolution of LLM investors’ beliefs as a function of the fundamental, and reveals three regimes in how LLM investors reason about coordination. For low θ (below $\underline{\theta}$), nearly all investors report “no belief”, consistent with redeeming being the dominant strategy and requiring no strategic reasoning. For high θ (above $\bar{\theta}$), “no belief” again dominates, now because staying is the dominant strategy.

In the intermediate region, LLM beliefs become heterogeneous. As θ rises within this region, the share of investors expecting coordination on the efficient equilibrium becomes

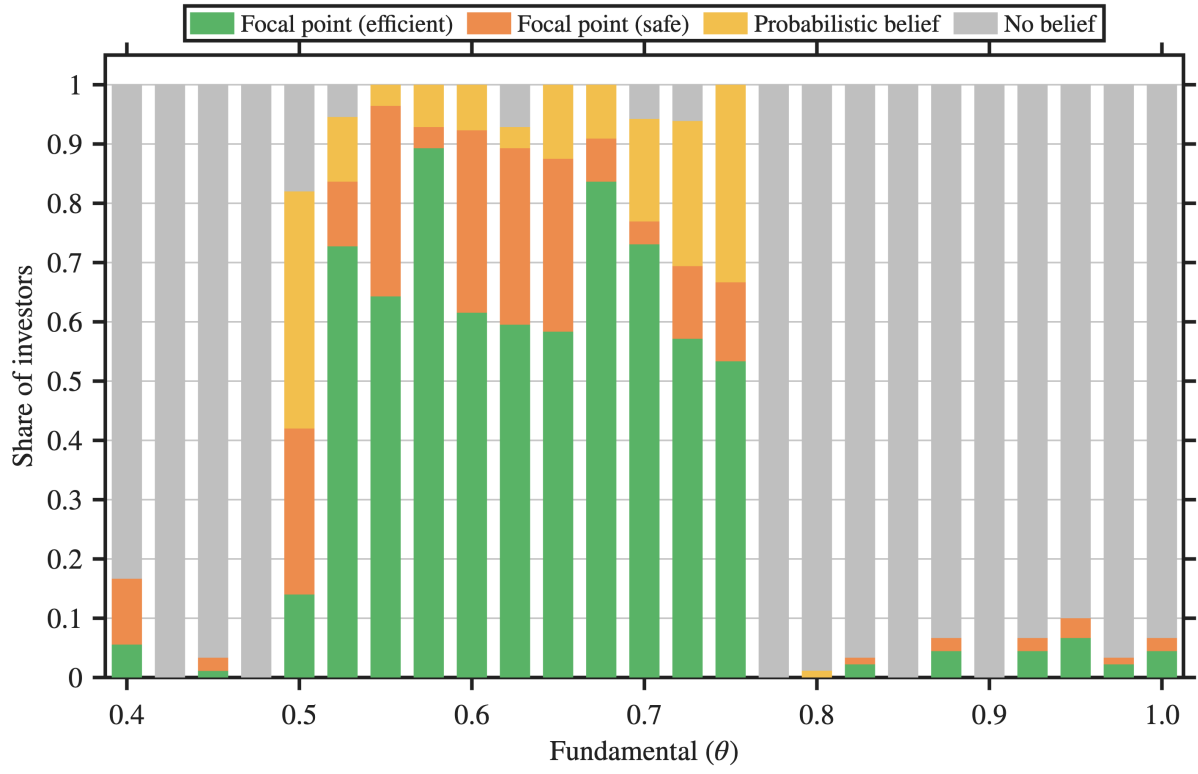


Figure 6: Evolution of LLM investors' beliefs about the behavior of other investors as a function of the fundamental θ .

Table 1: LLM investors’ critical cutoff estimates in the presence of default risk.

θ	Mean($\widehat{W}$)	Var($\widehat{W}$)	Theoretical ($\widehat{W}$)
0.50	0.000	0.000	0.000
0.55	13.556	1.432	13.333
0.60	20.000	0.000	20.000
0.65	24.000	0.000	24.000
0.70	26.667	0.000	26.667
0.75	28.571	0.000	28.571

The means and variances of the critical cutoff $\widehat{W}$ are inferred from the chain-of-thought for each individual LLM investor.

more prevalent. This aligns with the game’s structure: higher fundamentals increase the critical mass of redeemers required for staying to be optimal ($\widehat{W}$), enlarging the basin of attraction of the all-stay equilibrium. In other words, as the economic fundamental improves, it’s less risky to trust that others will stay, with the result that more of the LLMs converge on believing that others stay.

Interestingly, the agreement on $\widehat{W}$ holds despite disagreement on beliefs. As Table 1 shows, the cutoff $\widehat{W}$ inferred from the chain-of-thought, matches the theoretical values across the interval with only minor cross-investor variation. Put differently, LLM investors agree on where the threshold lies even where they disagree on what others will do.¹³

A natural concern is that the LLM is recalling a familiar solution to a redemption game rather than reasoning from the underlying economic structure. If so, the observed belief heterogeneity could simply reflect the framing of the problem. To address this concern, Appendix C considers three variations: (i) removing default risk, (ii) replacing the fund-redemption setting with a neutral action-choice problem, and (iii) expressing the same payoffs in natural language so that investors must infer returns from the scenario. Across all three variations, belief heterogeneity persists and remains concentrated in the intermediate region where the model admits multiple equilibria. This suggests that the result is driven by the strategic structure of the problem rather than by pattern matching to a familiar financial setting.

¹³This pattern indicates that LLM investors rely on simple expected value comparisons. It follows that default risk plays no role in LLM behavior not because such risk is immaterial in principle, but because LLM investors’ decision rules do not appear to incorporate concerns about risk or ambiguity. In Appendix C we report the equivalents of Figures 5 and 6 in the absence of default risk, confirming that LLM behavior is unchanged.

VI. Treatments

The two forms of misbehavior arise from different mechanisms: belief heterogeneity for LLM investors and learning bias for Q-learning investors. We therefore consider three remedies. The first two target belief heterogeneity among LLM investors, while the third targets the hot stove effect in Q-learning. In the first treatment, we replace common knowledge of θ with a noisy private signal for the investors. The standard global games logic suggests that this should resolve the indeterminacy in investors' beliefs and deliver a unique equilibrium. The second treatment instead involves providing LLM investors with a history of previous play that should work as an anchor to beliefs. Finally, to address the hot stove effect of Q-learning investors, we consider synchronous updating where the Q-value for both actions are updated after each episode.

A. Remedying belief heterogeneity with private signals

This remedy introduces fundamental uncertainty. Instead of directly observing θ , each investor i receives a private signal $\theta_i = \theta + \eta_i$ where the η_i are i.i.d. noise drawn from $[-\eta, +\eta]$. The parameter $\eta > 0$ is a measure of information dispersion.

With private signals the redemption game becomes a global game with a unique equilibrium in threshold strategies (Carlsson and Van Damme, 1993; Morris and Shin, 1998). In this equilibrium, each investor redeems if and only if their signal falls below a common cutoff $\theta^* \in (\underline{\theta}, \bar{\theta})$. Under this threshold strategy, for a given level of the fundamental, θ , the fraction of investors who redeem is a function $w^*(\theta, \theta^*)$ that declines smoothly in θ . Details on the derivations are provided in Appendix D.

We incorporate private signals by modifying the prompts for the LLM investors as shown in Prompt 2. The investors observe their private signal and also know the generic signal structure and distributions for the fundamental and noise term. They also are told that other investors receive private signals.

```

10 The fundamental value  $\theta$  (theta) of the fund is randomly drawn from
    the interval  $[0,1]$  but you do not directly observe it. All values
    of  $\theta$  are equally likely.
11 Instead you receive a private signal  $x_i$  defined as  $x_i = \theta + \epsilon_i$ ,
    where  $\epsilon_i$  is drawn uniformly from  $[-\eta, \eta]$  with  $\eta = \text{\%f}$ . Signals of
    different investors are drawn independently. Your private signal
    is  $x_i = \text{\%f}$ .
12 Do you choose to redeem your share or stay? State your decision

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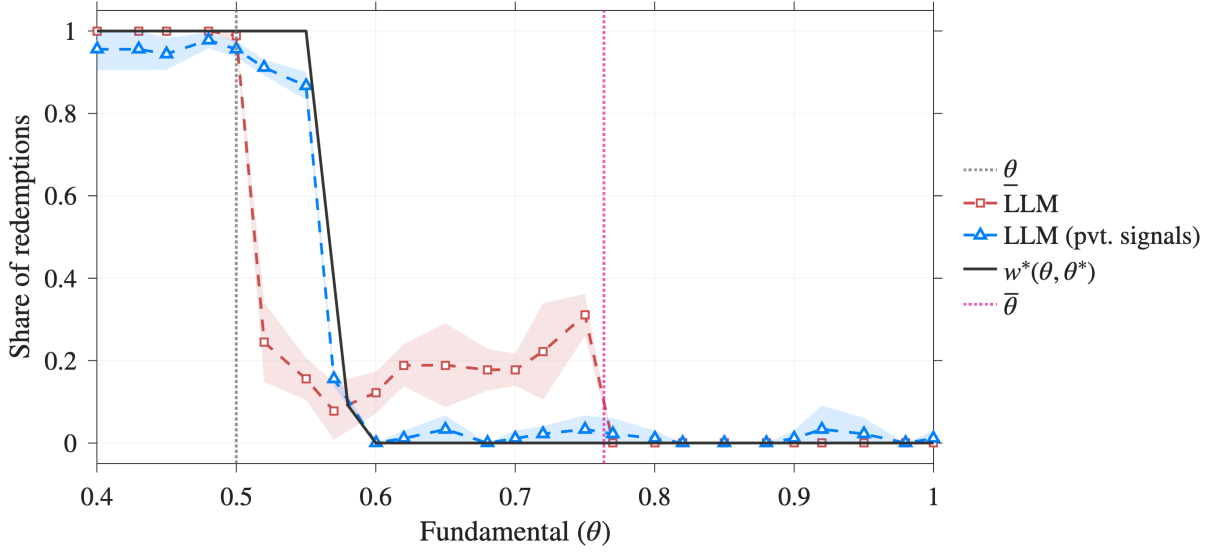


Figure 7: Share of redemptions as a function of the fundamental θ . The line labelled “LLM” refers to the baseline case with LLM investors; “LLM (pvt. signals)” refers to the case where LLM investors receive private signals about the fundamental, where the signal dispersion is $\eta = 0.01$; “ $w^*(\theta, \theta^*)$ ” is the theoretical prediction for the share of investors who redeem given the threshold θ^* and the current value of the fundamental. Shaded areas show standard deviations constructed across three independent inference rounds. The vertical lines $\underline{\theta} = 0.5$ and $\bar{\theta} = 0.764$ are the bounds defined in Proposition 1.

```
with exactly one word: ``redeem'' or ``stay'' using the XML tag <
decision>...</decision>.
```

Prompt 2: Introducing fundamental uncertainty: The baseline prompt is supplemented with additional information after line 9.

Figure 7 compares the simulated outcomes with LLM investors to the theoretical benchmark in the presence of fundamental uncertainty. Introducing private signals reduces the dispersion in investors’ actions, as all investors formulate and use threshold strategies. Moreover, the aggregate redemption profile closely tracks the theoretical prediction $w(\theta^*, \theta)$. To further investigate the underlying mechanism, Figure 8 presents the mapping of justifications to beliefs and redemption decisions for $\theta = 0.75$ and $\eta = 0.01$. The contrast with Figure 5 is striking. Whereas investors exhibited substantial heterogeneity in both justifications and beliefs without fundamental uncertainty, the introduction of noisy signals induces convergence to a single justification to belief pathway.

In sum, based on their private signals, which range between 0.74 and 0.76, and the

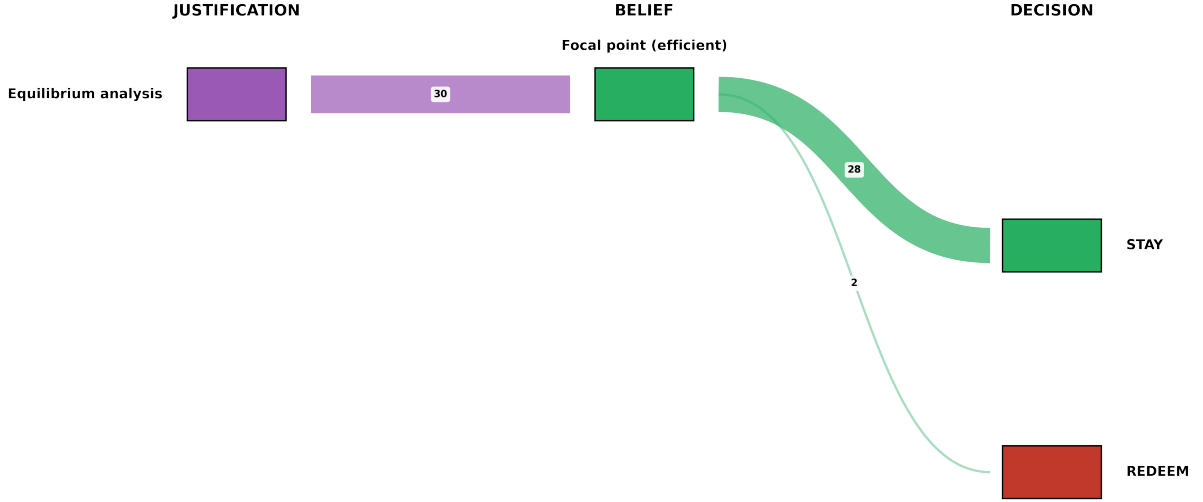


Figure 8: Mapping from justifications to beliefs and decisions where LLM investors receive private signals about the fundamental. The true level of the fundamental is $\theta = 0.75$ and the dispersion in signals is $\eta = 0.01$. The width of the links indicate investor counts.

calculated threshold, 28 investors choose to stay. The two redemptions (investors 7 and 29) are *not* driven by adverse signals or divergent beliefs. Both draw high signals ($\theta_7 = 0.745$, $\theta_{29} = 0.752$), both derive the same mean-field threshold, and both explicitly conclude that staying is optimal. Yet each is recorded as redeeming, an action inconsistent with its own derived conclusion. We read these two cases not as genuine redemption decisions but as computational execution errors following the audit of their reasoning text.

B. Remedying belief heterogeneity with revealed history

A second route to coordinating beliefs is to give investors a common record of past plays. Unlike private signals, history does not alter the game’s payoffs or provide information about the fundamental: the multiplicity of the baseline remains intact in principle. What history supplies is a shared, public basis for forming beliefs about how others will act. By the logic of inductive, case-based reasoning (Argenziano and Gilboa, 2012), a commonly observed record of previous outcomes lets players coordinate on the equilibrium that past play has established. History functions here not as data from which investors learn an unknown payoff (as payoffs are already known) but as a coordination device informing each investor what others have done, and hence what others are likely to do. This generates a prediction that distin-

guishes history from private signals. Whereas signals select a unique threshold equilibrium pinned down by the information structure, history selects whichever equilibrium past play has favored: coordination is inherited from the realized record rather than determined by theory.

To implement this, we extend the interaction from the three rounds of the baseline to twenty, holding the fundamental θ fixed throughout. The first ten rounds are played blind, i.e., each LLM investor decides without any record of past play, exactly as in the baseline. From the eleventh round onward, each LLM investor is instead shown the full history of the preceding rounds. This includes the complete profile of actions and realized payoffs in each round. This information is appended to the prompt, with the record updating after every subsequent round.¹⁴ Prompt 3 shows the appended block.

```

1 Previous rounds with this same  $\theta$  value:
2 Round 1: Actions=[...], Payoffs=[...]
3 :
4 Round 10: Actions=[...], Payoffs=[...]
```

Prompt 3: Revealing history. From the eleventh round onward, the baseline prompt is supplemented with the record of past play at the same fundamental.

Figure 9 reports the outcome for $\theta = 0.75$. The top panel shows the share of redeeming investors across rounds. Over the first ten rounds, played blind, behavior reproduces our baseline belief heterogeneity result. The investors fail to coordinate, and the redemption share sits at an intermediate level. From the eleventh round, once the history is revealed, the investors coordinate. The bottom panel shows the mechanism: the common record selects a single belief, and the population converges on the corresponding equilibrium, consistent with selection by induction over past play (Argenziano and Gilboa, 2012).

The direction of coordination is inherited from the revealed history. When the average redemption over the blind phase falls below the critical $\widehat{W}$, i.e., the number of redemptions from the indifference condition, investors coordinate on all stay, while when it falls above, they coordinate on all redeem. The cutoff $\widehat{W}$ thus determines each investor's best response and also selects the equilibrium after history is revealed. Since $\theta = 0.75$, default events are relatively infrequent. Hence, by revealing the history over ten blind rounds, a few unlucky default events are smoothed over and investors coordinate on an unambiguous history.

¹⁴The history reveals the full profile of actions and payoffs, not a summary statistic, so investors observe the actual distribution of play rather than an aggregate.

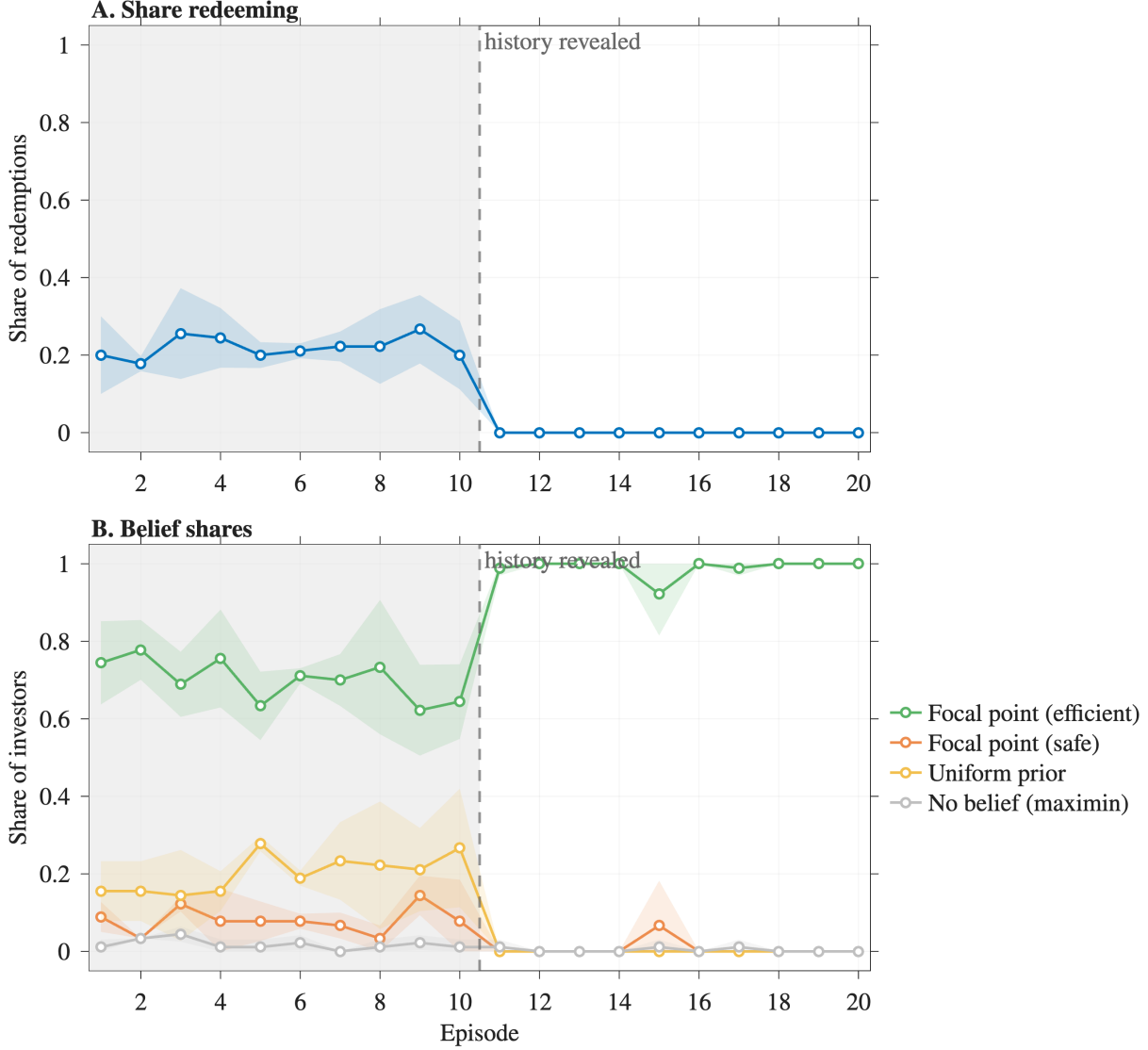


Figure 9: LLM investors with revealed history at $\theta = 0.75$. The top panel shows the share of redeeming investors across rounds. The bottom panel shows the share of investors holding each belief. For the first ten rounds investors play blind. From the eleventh round onward, the full record of past play is revealed. Shaded areas show standard deviations constructed across three independent inference rounds.

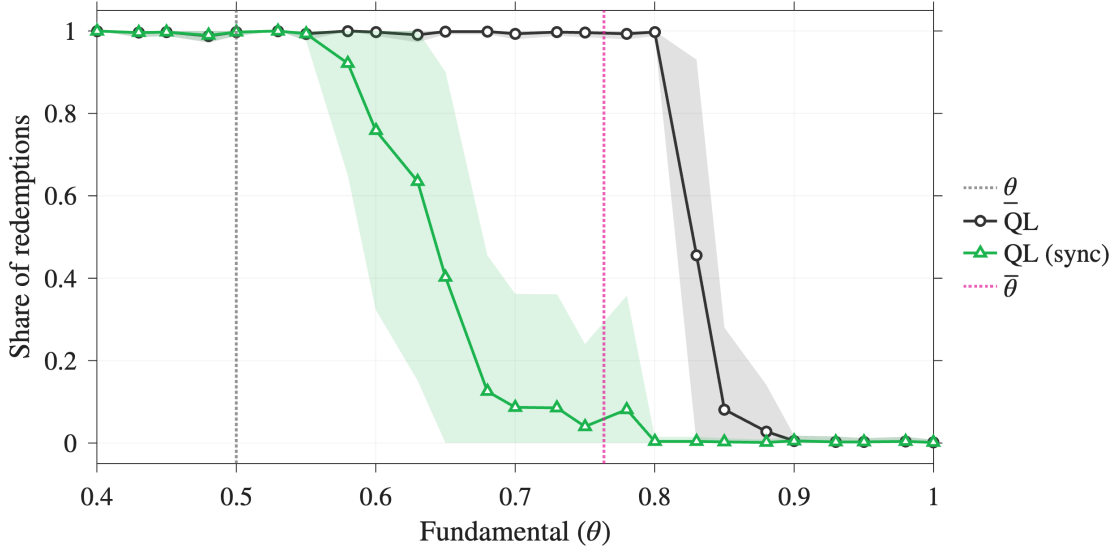


Figure 10: Share of redemptions as a function of the fundamental θ . The line labelled “QL” refers to the baseline case with Q-learning investors and asynchronous updating, while the line labelled “QL(sync)” represents the outcome using synchronous updating. Shaded areas show standard deviations constructed across twenty-five independent training rounds. The vertical lines $\underline{\theta} = 0.5$ and $\bar{\theta} = 0.764$ are the bounds defined in Proposition 1.

C. Treating the Q-learning hot stove effect with synchronous updating

The source of the hot stove effect in Proposition 2 was asynchronous updating, i.e., each Q-learning investor updates only the value of the action it takes. So, once a sequence of defaults drives the Q-value of stay below that of redeem, the investor switches to redeeming. Crucially, the Q-value of stay remains frozen and the error is never corrected.

The hot stove effect can be addressed by providing Q-learning investors with counterfactual feedbacks. In each episode an investor observes not only the realized payoff of the action it took in the previous episode, but also the payoff the unchosen action would have earned, and updates both action values accordingly. Following Banchio and Mantegazza (2025) we call this *synchronous updating*, since both Q-values are revised every period rather than only the value of the action played.

As Figure 10 shows, synchronous updating does treat the hot stove effect. But a side effect is that the variance of outcomes is significantly higher, suggesting that coordination has been impaired. Proposition 4 formalizes both these points.

Proposition 4: *Under synchronous updating, for any non-zero value of $\theta < 1$ and $\lambda < \frac{N-A+1}{N-1} (1 - \frac{1}{R})$, in the limit $t \rightarrow \infty$, the Q-value of stay is characterized by a stationary distribution with strictly positive variance. As a consequence, neither the all-stay nor the*

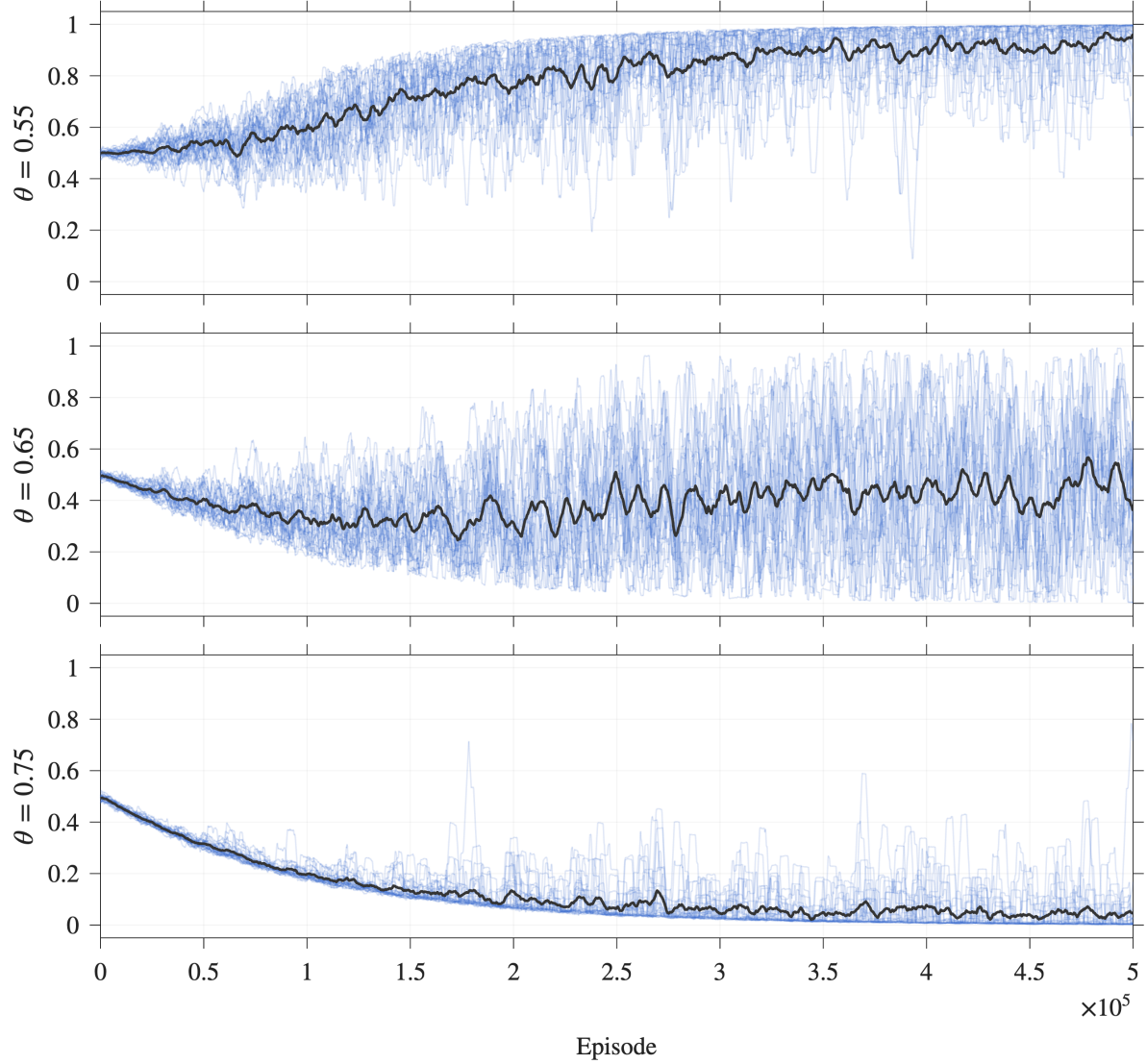


Figure 11: Evolution of the share of redemptions across episodes for three values of the fundamental (top panel $\theta = 0.55$; middle panel $\theta = 0.65$; bottom panel $\theta = 0.75$), with synchronous updating. Each blue line is the 5,000-episode rolling average of redemptions for one of the twenty five training rounds; the thick black line is the mean of these rolling averages across all twenty five rounds.

all-redeem equilibria are absorbing.

The intuition behind this result is as follows. Under synchronous updating, the Q-value for stay is updated every period regardless of the action taken. So it is an exponentially weighted moving average of the i.i.d. payoffs. It is characterized by a stationary non-degenerate distribution with positive variance as long as illiquidity is not too high. An important consequence of this is that the Q-learning investors never converge to either the all-stay or all-redeem equilibria. For example, if all investors find the all-redeem equilibrium, there is now a positive probability that the Q-value of stay for some investors will rise above R_1 and thus switch to stay. And this switch is guaranteed to happen in finite time. A similar argument applies when starting from the all-stay outcome.

Thus, while synchronous updating helps to mitigate the hot stove effect by ensuring that all-redeem is no longer an absorbing state, it still does not lead to an efficient outcome. In fact, the outcome is one where Q-learning investors never settle on either the all-stay and all-redeem equilibria, but instead exhibit persistent fluctuations. We see this clearly in Figure 11 which shows, for several different values of θ , how the share of redemptions evolves across episodes. In particular, for values of θ in the intermediate region, we see that across different runs the share of redemptions wildly fluctuates and never settles. But as θ increases the extent of these fluctuations decreases.

In the absence of default risk, the outcome obtained continues to be governed by Proposition 3. This is because the Q-value for redeem is fixed at R_1 and so what drives the outcome is how the Q-value for staying evolves. In the regions where staying is a strictly dominant action, synchronous learning ensures that this is the unique learned outcome. While in the region with multiple equilibria, and insofar as learning converges while the rate of exploration is high (i.e., $-\log\beta \ll \alpha$), investors learn to best-respond as if other investors randomize between staying and redeeming. Thus, synchronous updating does not influence the exploration effect.

Our result complements those Banchio and Mantegazza (2025). They show that in any supermodular game with a unique equilibrium, counterfactual Q-learners learn to play the Nash equilibrium. We show that with default risk this need not be the case. This is because the stationary distribution for the stay payoff has strictly positive variance. As a consequence, there is a strictly positive probability of deviating from an equilibrium outcome. This is also the case in the absence of default risk, when the exploration effect continues to persist, thus leading to an outcome, which is consistent with level-1 reasoning.

VII. Conclusions

We study how AI agents behave in settings where outcomes depend not only on economic fundamentals but also on the strategic interaction among them. In these settings, the same economic fundamental is consistent with multiple possible outcomes, and financial fragility emerges as the result of agents’ coordination failure. We use a mutual fund redemption game as a canonical example of a framework with these characteristics. Fund redemptions depend on the state of the economy, but also on the number of investors who redeem. Holding payoffs, information, and incentives fixed, we find that AI architecture alone select aggregate outcomes. Q-learning agents coordinate successfully, but frequently on the wrong outcome: they redeem even when fundamentals are strong. LLM agents redeem less overall, but do not converge on a common action because they form different beliefs about what others will do.

The broader lesson is that AI architecture is not just an implementation choice: even with the same fundamentals, payoffs, and information, different systems can produce different (and often inefficient) aggregate outcomes. Investors and institutions, therefore, need to understand that their choices of AI tools do shape financial decisions and aggregate outcomes.

For policymakers, the key implication is that financial stability may depend not only on financial incentives, but also on which AI systems investors and institutions adopt. For example, investor protection tools (such as MiFID suitability questionnaires) could assess not only financial literacy, but also AI literacy. Moreover, stress tests could examine how different AI architectures behave under pressure, while disclosure rules could require institutions to explain how those systems shape investor behavior.

Market safeguards should also work across architectures. Circuit breakers, trading halts, and related measures can limit destabilizing feedback loops (Breedon, 2026), especially when AI agents react rapidly and in correlated ways. Traditional prudential tools remain useful, but they may need to be supplemented by safeguards designed for the speed and scale at which AI agents trade.

More broadly, AI coordination failures may arise wherever strategic interaction among machines matters, including autonomous vehicle systems, supply chains, and resource allocation. Traditional safeguards may therefore need to be complemented by tools that monitor and stress-test interactions among AI agents. Even when each agent is aligned with its own objective, groups of agents can still produce inefficient outcomes (Lorè and Heydari, 2024). Our findings highlight the importance of multi-agent alignment in addition to individual system alignment.

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Appendix A. Proofs

Proof of Lemma 1. For extreme values of the fundamental θ , investors have a dominant action. We start with the low values of θ . Irrespective of what other investors choose, redeeming at $t = 1$ is optimal for an investor when even the highest payoff they can accrue at $t = 2$, which corresponds to the payoff they obtain if no other investors redeem, is smaller than R_1 . Formally, this is the case when $R_1 R \theta < R_1$, that is, when $\theta < \underline{\theta} \equiv \frac{1}{R}$.

Symmetrically, staying until $t = 2$ is a dominant action when even the worst payoff that an investor expects to receive at the final date, which corresponds to the payoff accrued if all $A - 1$ investors redeem, is larger than R_1 . Formally, this is the case when $R_1 R \theta^{\frac{N-(A-1)(1+\lambda)}{N-(A-1)}} > R_1$ that is when $\theta > \bar{\theta} = \frac{1}{R} \frac{N-(A-1)}{N-(A-1)(1+\lambda)}$.

When $\theta \in [\underline{\theta}, \bar{\theta}]$, both redeeming at $t = 1$ and not redeeming are equilibria. If an investor expects others to redeem, it is optimal for them to redeem as well, since staying until $t = 2$ would yield a lower payoff than R_1 . Conversely, if no one else redeems at $t = 1$, it is optimal not to redeem either, as the payoff from staying, $R_1 R \theta$, exceeds R_1 for all $\theta > \underline{\theta}$. This completes the proof. $\square$

Proof of Proposition 1. Since the total returns accrue to the investors, utilitarian welfare is defined by

$$U(W, \theta) = W R_1 + \theta R R_1 (N - W(1 + \lambda)), \quad (\text{A.1})$$

which is the sum of the payoff to the W investors who redeem plus the expected returns for those $N - W$ investors who stay, which include the non-active investors. The cost of servicing redemptions is accounted for with the additional λ term.

An unconstrained planner chooses $W \in [0, A]$ to maximize (A.1). Since $U(W, \theta)$ is linear in W , the solution is a corner, determined by the sign of

$$\frac{\partial U}{\partial W} = R_1 [1 - \theta R(1 + \lambda)]. \quad (\text{A.2})$$

Hence $W^{FB} = A$ if $\theta < \theta^{FB}$, and $W^{FB} = 0$ if $\theta > \theta^{FB}$, where

$$\theta^{FB} \equiv \frac{1}{R(1 + \lambda)}. \quad (\text{A.3})$$

A constrained planner can only select among outcomes that are themselves equilibria of the redemption game, i.e., those characterized in Lemma 1. This means that for $\theta < \underline{\theta}$, the only implementable equilibrium is $W^* = A$. Similarly, when $\theta > \bar{\theta}$, the only implementable equilibrium is $W^* = 0$. In the intermediate region, both equilibria are implementable, with $U(0, \theta) > U(A, \theta)$. Consequently, the planner chooses $W^* = 0$. $\square$

Proof of Proposition 2. We prove the result for a single Q-learning investor and a fixed fundamental $\theta < 1$. The results directly extend to all states since Q-values are updated state-by-state. We discuss how the result extends to A investors at the end.

In what follows, we suppress the investor index and the state argument in the Q-values. Let $Q_t(a)$ be the Q-value of action $a \in \{a_R, a_S\}$ in episode t and $\pi_t(a)$ the associated realized

reward. In this notation, under asynchronous updating,

$$Q_{t+1}(a) = (1 - \alpha)Q_t(a) + \alpha \pi_t(a), \quad (\text{A.4})$$

where

$$\pi_t(a_R) = R_1, \quad \text{and} \quad \pi_t(a_S) = Z_t B(W), \quad (\text{A.5})$$

with $B(W) = R R_1 \phi(W)$, where $\phi(W) = (N - W(1 + \lambda))/(N - W)$ and $Z_t \in \{0, 1\}$ is the default indicator with $\Pr(Z_t = 0) = 1 - \theta$, drawn independently across dates. Exploration follows $\varepsilon_t = \beta^t$ and decays at rate $-\log \beta$.

Bounding the Q-values. Each Q-value stays within the interval spanned by its initialization and the range of its rewards, since the update is a convex combination of the previous value and a new reward. For the redeem value, the reward is the constant R_1 , so $Q_{t+1}(a_R) = (1 - \alpha)Q_t(a_R) + \alpha R_1$ is a convex combination of $Q_t(a_R)$ and R_1 . By induction,

$$Q_t(a_R) \geq \underline{r} := \min\{Q_0(a_R), R_1\} \quad \text{for all } t. \quad (\text{A.6})$$

For the stay value, the reward satisfies $0 \leq \pi_t(a_S) \leq \bar{B}$ with $\bar{B} := R R_1$ (the best case $W = 0, Z_t = 1$), so $Q_{t+1}(a_S)$ is a convex combination of $Q_t(a_S)$ and a quantity at most $\bar{B}$. By induction,

$$Q_t(a_S) \leq \bar{Q} := \max\{Q_0(a_S), \bar{B}\} \quad \text{for all } t. \quad (\text{A.7})$$

Since $\underline{r} < \bar{Q}$ in general, staying may be preferred initially. The question is whether a run of defaults can overturn this permanently.

A finite run of defaults locks in redemption. If the investor stays and the fund defaults ($Z_t = 0$), then $\pi_t(a_S) = 0$ and the stay value contracts geometrically, $Q_{t+1}(a_S) = (1 - \alpha)Q_t(a_S)$. Iterating over K consecutive defaults and bounding the starting value by the ceiling $\bar{Q}$ gives $Q_t(a_S) \leq (1 - \alpha)^K \bar{Q}$. The condition $(1 - \alpha)^K \bar{Q} < \underline{r}$ is equivalent, taking logarithms and noting $\log(1 - \alpha) < 0$, to $K > \log(\underline{r}/\bar{Q})/\log(1 - \alpha)$; hence any run of at least

$$K^* := \left\lfloor \frac{\log(\underline{r}/\bar{Q})}{\log(1 - \alpha)} \right\rfloor + 1 \quad (\text{A.8})$$

consecutive defaults drives $Q_t(a_S) < \underline{r} \leq Q_t(a_R)$. Once this happens, the investor switches to redeem. Assuming that the investor only plays the greedy action henceforth, then in all subsequent episodes and under asynchronous updating only $Q_t(a_R)$ is updated while $Q_t(a_S)$ is left unchanged. Since redeeming pays the constant R_1 , the value $Q_t(a_R)$ remains a convex combination of quantities at least $\underline{r}$ and so stays at or above $\underline{r} > Q_t(a_S)$. The greedy action therefore remains a_R forever. It remains to show that the investor eventually becomes permanently greedy and that a run of K^* defaults does occur.

Exploration ceases in finite time. The exploration probabilities are summable, $\sum_{t \geq 0} \varepsilon_t = \sum_{t \geq 0} \beta^t = 1/(1 - \beta) < \infty$. So, by the first Borel-Cantelli lemma, the investor only explores finitely often. Formally, there almost surely exists a date $T_E < \infty$, such that for all $t \geq T_E$

the investor always plays the greedy action.

A lock-in run is inevitable. From T_E onward the investor always plays the greedy action. If the investor redeems at some $t \geq T_E$, then by the absorption argument above, its stay value is frozen while its redeem value stays at or above $\underline{r}$, so it redeems forever and we are done. If, however, the investor chooses to stay, then for each $t \geq T_E$, the reward carries an independent default draw with $\Pr(Z_t = 0) = 1 - \theta > 0$. Partition these stay-dates into consecutive disjoint blocks of size K^* . A block is *bad* if all K^* of its draws default. The probability of this event is $(1 - \theta)^{K^*} > 0$. The probability that none of the first J blocks is bad is $(1 - (1 - \theta)^{K^*})^J \rightarrow 0$ as $J \rightarrow \infty$. So almost surely some block consists entirely of defaults. Thus, following this run of K^* consecutive defaults while choosing to stay drives $Q_t(a_S) < \underline{r} \leq Q_t(a_R)$ and locks in redemption from that date on.

Extension to A investors. The argument above conditions only on a single investor's own default draws and own actions. Three observations extend the conclusion to the full system. First, the ceiling invariant is unaffected: $B_t \leq \bar{B} = R R_1$ for every $W \geq 0$, so $\bar{Q}$ remains a valid upper bound on each investor's stay value whatever the others do. Second, the contraction-on-default $Q_{t+1}(a_S) = (1 - \alpha)Q_t(a_S)$ and the count K^* in (A.8) depend on no other investor, so the absorption mechanism operates investor-by-investor. Third, defaults occur whenever the realized return is zero, an event of probability $1 - \theta > 0$ at every date for every investor as long as $\theta < 1$. Hence, conditional on any action path of the others, each investor almost surely experiences a run of K^* defaults while staying and greedy, and locks into redemption. Applying this to each investor in turn, the whole system reaches all-redeem in almost surely finite time.

Strategic complementarity tightens the result rather than weakening it. The stay reward B_t is decreasing in W : as other investors lock into redemption and W rises, the realized stay value of any remaining investor contracts from a lower level, so the threshold $\underline{r}$ is crossed after fewer defaults. Lock-in is therefore reached weakly earlier than in the single-investor benchmark. $\square$

Proof of Proposition 3. We characterize the dynamics in the absence of default risk under a mean-field approximation where we replace each investor's stochastic update rule with its conditional expectation and summarize the actions of the other symmetric investors through the redemption probability, p . The approximation is exact in the limit of a large population. For finite α and A , while the threshold remains $\tilde{\theta}$, the transition is smooth around the threshold.

We first start with some preliminary identities before presenting the proof. Throughout, we fix the fundamental $\theta < 1$.

The redemption probability. Let $\Delta_t = Q_t(a_R) - Q_t(a_S)$ indicate that Q-value difference between redeem and stay. Under uniform exploration, the probability that an investor redeems is

$$p_t = \frac{1}{2}\varepsilon_t + (1 - \varepsilon_t)\mathcal{H}(\Delta_t), \quad (\text{A.9})$$

where $\mathcal{H}(\cdot)$ is a heaviside function that returns one when its argument is non-negative, and zero otherwise. The redemption probability is a convex combination of the exploration value $\frac{1}{2}$ and the current greedy action.

Expected payoffs. Denote the expected stay payoff by

$$V_S(\theta, p) \equiv \theta \mathbb{E}_{W \sim \text{Bin}(A-1, p)}[B(W)] = \theta R R_1 \Phi(p), \quad (\text{A.10})$$

where

$$\Phi(p) \equiv \mathbb{E}_{W \sim \text{Bin}(A-1, p)}[\phi(W)] \quad (\text{A.11})$$

is the expected discount from the mutual fund disposing of assets to service redemptions. The expected payoff from redeeming is simply $V_R = R_1$.

Rate at which the fundamental is revisited. The fundamental is drawn i.i.d. across episodes. The Q-value associated with a fundamental is therefore revised in a given episode with uniform probability ν . The effective learning gain per episode is accordingly $\alpha\nu$, whereas exploration decays at rate $-\log \beta$ per episode.

Separation of timescales. The proof rests on the idea that there is a separation between the learning timescale, i.e., how fast the Q-values converge, and how fast exploration decays. The key assumption is that

$$-\log \beta \ll \alpha/\nu, \quad (\text{A.12})$$

which implies that learning is relatively faster than the decay of exploration. Under this assumption there is a time window where

$$\frac{1}{\alpha/\nu} \ll t \ll \frac{1}{-\log \beta}. \quad (\text{A.13})$$

We anchor this interval by the geometric mean of its end-points:

$$T_\star \equiv \frac{1}{\sqrt{\frac{\alpha}{\nu}(-\log \beta)}}, \quad (\text{A.14})$$

such that

$$\alpha T_\star = \sqrt{\frac{\alpha/\nu}{-\log \beta}} \gg 1, \quad (-\log \beta) T_\star = \sqrt{\frac{-\log \beta}{\alpha/\nu}} \ll 1. \quad (\text{A.15})$$

The first bound ensures that initial conditions are washed out, while the second ensures exploration barely decays while learning converges.

Step 1: Mean-field updating. The first step is to replace the reward from staying with the expected payoff, i.e., $\mathbb{E}[\pi_t(a_S)] \rightarrow V_S(\theta, p)$. Under asynchronous updating, stay is sampled with probability $1 - p_t$, while redeem is sampled with probability p_t . So, the conditional mean

updated Q-values are given by

$$\mathbb{E}[Q_{t+1}(a_S) \mid Q_t] = (1 - \frac{\alpha}{\nu}(1 - p_t))Q_t(a_S) + \alpha(1 - p_t)V_S(\theta, p_t) \quad (\text{A.16})$$

$$\mathbb{E}[Q_{t+1}(a_R) \mid Q_t] = (1 - \frac{\alpha}{\nu}p_t)Q_t(a_R) + \alpha p_t R_1. \quad (\text{A.17})$$

Step 2: Initial conditions are irrelevant. Unrolling $Q_t(a_S)$ in the equation above, we get

$$Q_t(a_S) = \left[\prod_{k < t} \left(1 - \frac{\alpha}{\nu}(1 - p_k)\right) \right] Q_0(a_S) + \frac{\alpha}{\nu} \sum_{k < t} (1 - p_k) \left[\prod_{k < j < t} \left(1 - \frac{\alpha}{\nu}(1 - p_j)\right) \right] V_S(\theta, p_k). \quad (\text{A.18})$$

The first term in Equation (A.18) captures the impact of the initial conditions. As long as the stay probability satisfies $1 - p_t \geq c > 0$, i.e., it never converges to zero, then $1 - \frac{\alpha}{\nu}(1 - p_t) < 1 - \frac{\alpha}{\nu}c < 1$. This implies that $\prod_{k < t} (1 - \frac{\alpha}{\nu}(1 - p_k)) \leq (1 - \frac{\alpha}{\nu}c)^t = \exp\{t \log(1 - \frac{\alpha}{\nu}c)\} < \exp\{-\frac{\alpha}{\nu}ct\}$. Evaluated at T_* , the fact that $\frac{\alpha}{\nu}T_* \gg 1$ implies $\exp\{-\frac{\alpha}{\nu}cT_*\} \rightarrow 0$ and so initial conditions have no bearing on the outcome (we establish what c is in step 4). The second term in Equation (A.18) is a weighted average of past targets $V_S(\theta, p_k)$. In step 4, we show these are constant up to a small error.

Step 3: Aggregating to the population level. We now turn our attention to the redemption probability where we can express

$$\Delta(\theta, p) = V_R - V_S(\theta, p) = R_1 - \theta R R_1 \Phi(p). \quad (\text{A.19})$$

Substituting this into the redemption probability allows us to represent the dynamics by the mapping

$$p_{t+1} = \frac{1}{2}\varepsilon_t + (1 - \varepsilon_t)\mathcal{H}(\Delta(\theta, p_t)). \quad (\text{A.20})$$

Step 4: Exploration pins the population at equal mixing. Subtracting $\frac{1}{2}$ from either side of Equation (A.20) yields

$$p_{t+1} - \frac{1}{2} = (1 - \varepsilon_t) \left(\mathcal{H}(\Delta_t(\theta, p_t)) - \frac{1}{2} \right). \quad (\text{A.21})$$

Since $\mathcal{H}(\Delta_t(\theta, p_t)) \in \{0, 1\}$, this implies that $|\mathcal{H}(\Delta_t(\theta, p_t)) - \frac{1}{2}| = \frac{1}{2}$ irrespective of $\Delta_t(\theta, p_t)$. It thus follows that

$$\left| p_{t+1} - \frac{1}{2} \right| = \frac{1}{2}(1 - \varepsilon_t) = \frac{1}{2}(1 - \beta^t) = \frac{1}{2}(1 - e^{t \log \beta}) \leq \frac{1}{2}t(-\log \beta). \quad (\text{A.22})$$

Because this bound holds irrespective of the Q-gap, p_t is pinned down entirely from the exploration schedule. Evaluating the above inequality for $t \leq T_*$, we get

$$\left| p_{t+1} - \frac{1}{2} \right| \leq \frac{1}{2}(-\log \beta)T_* = \frac{1}{2}\sqrt{-\log \beta/(\alpha/\nu)} \ll 1. \quad (\text{A.23})$$

This establishes that $p_k = \frac{1}{2} + O(\sqrt{-\log \beta / (\alpha/\nu)})$ with two consequences. First, it secures the constant required in step 2: $1 - p_k \geq \frac{1}{2} - \frac{1}{2}\sqrt{-\log \beta / (\alpha/\nu)} \geq \frac{1}{4}$, so we may take $c = \frac{1}{4}$. Second, the target is a constant up to the same small error,

$$|V_S(\theta, p_k) - V_S(\theta, \frac{1}{2})| \leq O\left(\sqrt{\frac{-\log \beta}{\alpha/\nu}}\right). \quad (\text{A.24})$$

So the second term in Equation (A.18) converges to the equal-mixing expected stay payoff $V_S(\theta, \frac{1}{2})$. In the limit $-\log \beta / (\alpha/\nu) \rightarrow 0$, the difference goes to zero and the value is exact.

Step 5: Deriving the threshold. At the end of the reference window, the Q-gap is therefore evaluated at $p = \frac{1}{2}$. The greedy action is redeem if and only if

$$R_1 > \theta R R_1 \Phi(\frac{1}{2}). \quad (\text{A.25})$$

The threshold is the fundamental at which the two expected payoffs coincide, $V_S(\tilde{\theta}, \frac{1}{2}) = V_R$, i.e., $\tilde{\theta} R R_1 \Phi(\frac{1}{2}) = R_1$. The payoff R_1 cancels from both sides, leaving $\tilde{\theta} = \frac{1}{R \Phi(\frac{1}{2})}$. For $\theta < \tilde{\theta}$ the greedy action at T_* is redeem, while for $\theta > \tilde{\theta}$ it is stay.

Step 6: The selected action persists as exploration fades. Steps 1-5 fix the greedy action at T_* . We now check it does not reverse once the investor stops exploring. Take $\theta > \tilde{\theta}$, so stay is greedy at T_* . After the window the redeem propensity falls from $\frac{1}{2}$ toward 0 since the greedy action is stay and $\varepsilon_t \rightarrow 0$. The mean value for stay reaches a maximum $R_1 R \theta \Phi(0)$, while the mean value for redeem stays at R_1 . Since the stay value was already above the redeem value at T_* , the gap has only gotten wider and hence stay remains the greedy choice and the population settles into all-stay. The case $\theta < \tilde{\theta}$ is symmetric, with $p_t \uparrow 1$, the stay target falling, and all-redeem selected. $\square$

Proof of Proposition 4. In this proof, we recycle some of the notation introduced in the proof of Proposition 2.

Under *synchronous* updating both action-values are revised every period, using the realized payoff of the action taken and the counterfactual payoff of the action not taken. We first establish the stationary law of the stay Q-value at a fixed number of redemptions, W (Step 1), and then use it to show neither corners are absorbing (Steps 2–3).

Step 1: Stationary law of the stay-value. Because synchronous updating revises the stay-value every period regardless of the action taken, and the redemption count is held at W , the stay-value evolves as

$$Q_{t+1}(a_S) = (1 - \alpha)Q_t(a_S) + \alpha Z_t B(W), \quad (\text{A.26})$$

which is an exponentially-weighted moving average of the i.i.d. stay payoffs. Iterating,

$$Q_t(a_S) = (1 - \alpha)^t Q_0(a_S) + \alpha \sum_{k=1}^t (1 - \alpha)^{t-k} Z_k B(W). \quad (\text{A.27})$$

The first term $(1 - \alpha)^t Q_0(a_S)$ becomes vanishingly small. So, in the stationary limit, the Q-value, $Q_\infty(a_S)$, is bounded in $[0, B(W)]$.

Define $p \equiv \text{Prob}(Q_\infty(a_S) > R_1)$. In the subsequent steps, we require that $p > 0$. This, in turn, requires that $B(W) > R_1$. A sufficient condition for this is that $\lambda < \frac{N-A+1}{A-1} (1 - \frac{1}{R})$.

Step 2: All-redeem is not absorbing. Consider the population at the all-redeem configuration, so each investor faces $W = A - 1$ co-redeemers and stay payoff $B(A - 1) > 0$. Under *asynchronous* updating this configuration is absorbing. Synchronous updating removes this. Because the stay-value is updated through the counterfactual even while the investor redeems, it converges in law to $Q_\infty(a_S)$ of Step 1 with $W = A - 1$, dispersed around $\theta B(A - 1)$.

Because each investor has a probability $p > 0$ of switching to stay, it immediately follows that the all-redeem configuration is not absorbing.

Step 3: All-stay is not absorbing. The argument is symmetric. At all-stay every investor faces $W = 0$ and stay payoff $B(0) = R R_1$. The stay-value converges in law to $Q_\infty(a_S)$ with $W = 0$, dispersed around $\theta B(0)$. Let $q \equiv \Pr(Q_\infty(a_S) < R_1)$; by Step 1, $q > 0$ for all $\theta < 1$. An investor whose stay-value falls below R_1 has greedy action a_R and redeems. The probability that at least one of the A investors redeems in a given episode is at least $1 - (1 - q)^A > 0$, and by the same Borel–Cantelli argument a redemption occurs almost surely in finite time. All-stay is therefore not absorbing either. $\square$

Appendix B. Prompts for justification and belief extraction

In this appendix, we provide the prompts used to extract beliefs and reasoning from the chains of thoughts of the LLM investors. There are two prompts: one to extract the beliefs of the LLM investors (Prompt B.1) and the second to cluster them into canonical game-theoretic categories (Prompt B.2).

```
1 You are analyzing an investor's reasoning in a coordination game (
  mutual fund redemption).
2 CONTEXT:
3 The investor was given this prompt:
4 \{investor\_prompt\}
5 Their FINAL DECISION was: \{decision\}.
6 TASK:
7 The investor's reasoning trace may contain multiple considerations,
  calculations, and hypothetical beliefs. Your job is to identify:
8   1. The DECISIVE BELIEF -- the specific belief about other
     investors' behavior that ultimately determined their final
     decision
9   2. The JUSTIFICATION -- the reasoning that led them to form this
     belief
10  3. The CAUSAL CHAIN from justification -> belief -> decision
11 Note: The investor might explore multiple scenarios (e.g., "if
     others redeem with p=0.5..." or "in the all-stay equilibrium...")
     but ultimately commits to ONE belief that drives their final
     action. Find that committed belief.
12 REASONING TRACE:
13 \{explanation\}.
14 INSTRUCTIONS:
15   1. Read the ENTIRE reasoning trace
16   2. Identify where the investor COMMITS to a specific belief
     about others
17   3. Identify what JUSTIFICATION led them to form this belief
18   4. Extract the exact text passage that reveals the belief
     commitment
19   5. Construct the causal DAG
```

Prompt B.1: Prompt to extract justifications and beliefs of the LLM investors.

```
1 You are analyzing causal reasoning patterns extracted from 30
  investors.
2 Below are the causal DAGs (justification -> belief -> decision) for
  each investor, described in free-form text. Each entry also
  includes:
```

```

3     - evidence\_quote: The exact text from the reasoning trace
      supporting the extraction
4     - confidence: How clearly the causal chain was stated (HIGH/
      MEDIUM/LOW)
5 Your task is to:
6     1. Identify the distinct categories of JUSTIFICATIONS present in
      the data (use EXACTLY 3-4 categories)
7     2. Identify the distinct categories of BELIEFS present in the
      data
8     3. Assign each investor to one justification category and one
      belief category
9 IMPORTANT: Be PARSIMONIOUS with justification categories. Use only
      3-4 high-level categories. Merge similar reasoning patterns.
10 Reference list of standard JUSTIFICATION categories (consolidate
      into these):
11     - Payoff comparison: Reasoning based on comparing payoffs across
      equilibria (includes Pareto efficiency, strong fundamentals,
      threshold calculations)
12     - Strategic uncertainty: Reasoning based on uncertainty about
      others' behavior (includes uninformative priors, no
      information)
13     - Worst-case avoidance: Focused on guaranteed minimum payoff, no
      belief about others (maximin reasoning)
14     - Equilibrium analysis: Formal equilibrium concepts (includes
      mixed equilibrium analysis, deviation loss comparison)
15 Reference list of standard BELIEF categories (use these if they fit)
      :
16     - Focal point (efficient): Believe others coordinate on Pareto-
      efficient equilibrium (all-stay)
17     - Focal point (safe): Believe others coordinate on risk-dominant
      equilibrium (all-redeem)
18     - Uniform prior: Assume equal probability of others' actions, no
      specific expectation
19     - Mixed-strategy equilibrium: Believe others play the symmetric
      mixed equilibrium probability
20     - No belief: No probabilistic belief formed (used with worst-
      case/maximin reasoning)
21 INVESTOR DATA:
22 \{investor\_data\}

```

Prompt B.2: Prompt to cluster the extracted justifications and beliefs of the LLM investors into canonical categories.

Appendix C. Alternative prompts

In this appendix, we consider three alternative prompt structures. The first removes default risk and replaces the payoff from staying with the expected payoff from the game. Prompt C.1 shows the modification made to the baseline prompt. The second and third prompts retain the no-default-risk formulation but vary the framing. Prompt C.2 is a *neutral context prompt* that removes the fund-redemption framing entirely, while Prompt C.3 is a *narrative prompt* that preserves the fund-redemption context but presents the information in natural language, requiring investors to compute the payoff of each action from the scenario description. We explore these settings to rule out that LLM investors’ redemption decisions reflect pre-trained knowledge of a particular class of strategic games rather than genuine reasoning about economic fundamentals and strategic uncertainty.

Figure C.1 shows the share of redemptions as a function of the fundamental under the three proposed alternatives and the baseline case with default risk. The results are all qualitatively similar. All LLM investors correctly redeem in the full-redemption region and stay in the no-redemption region, irrespective of the prompt. In the intermediate region, the share of redemptions is decreasing in the fundamental as in the baseline case.

```

3 If you do not redeem your share, i.e., you stay, then you earn
  fraction * R1 * R * \theta, where:

```

Prompt C.1: No default risk: The baseline prompt is modified on line 3.

```

1 You are one of A active participants in a group of size N. Each
  participant chooses between two actions. Your objective is to
  maximize your payoff.
2 PAYOFF STRUCTURE:
3 If you choose Action A: your payoff is R1
4 If you choose Action B: your payoff is  $\theta R(N - W(1 + \lambda))/(N - W)$ , where:
5 -  $W \leq A - 1$  is the number of other active participants who choose
  Action A (the remaining  $\{N - A\}$  participants are passive and
  always choose Action B)
6 -  $\lambda$  is a parameter affecting how costly it is when more participants
  choose Action A
7 -  $\theta$  reflects the underlying state of the environment
8 -  $R$  is the baseline payoff multiplier
9 Which action do you choose? Answer using: <decision>A</decision> or
  <decision>B</decision>

```

Prompt C.2: Neutral context prompt.

- 1 You and $\{N-1\}$ other investors have each invested $\text{\textdollar}100,000$ in a mutual fund and have received shares. The fund invests the total amount raised based on its own investment strategy in a portfolio of assets.
- 2 The current value of the fund's assets is $\text{\textdollar}\{total_return_today:.2f\}$.
- 3 TODAY, you have to decide what to do with your shares.
- 4 If you REDEEM your shares today, you are guaranteed to receive $\text{\textdollar}\{individual_return_today:.2f\}$.
- 5 If you STAY invested, you will be repaid only after three months, and the amount you receive will depend on the value of the fund's assets at that time.
- 6 Based on current information about the fund and its strategy, if all investors remain invested, the value of the fund's assets is estimated to be $\text{\textdollar}\{future_return:.2f\}$ after three months. So, each investor will receive $\text{\textdollar}\{future_return_per_investor:.2f\}$.
- 7 But, if some investors redeem today, the fund's asset value after three months will be strictly lower than $\text{\textdollar}\{future_return:.2f\}$, because fewer assets remain invested. The fund services redemptions by selling assets at a discount due to illiquid market conditions: to generate $\text{\textdollar}50,000$ of cash today, for example, the fund has to sell $\text{\textdollar}\{liquidity_cost_example:.2f\}$ worth of assets, which eats into the fund's assets. The more investors redeem today, the larger the impact on the fund's assets for those who stay invested.
- 8 Out of a total of $\{N\}$ investors, a number $\{A_bar\}$, which includes yourself, are active and in a position to decide today whether to redeem their shares or stay invested; the others will stay invested regardless.
- 9 You aim to maximize the repayment you receive based on the above information.

Prompt C.3: Narrative prompt.

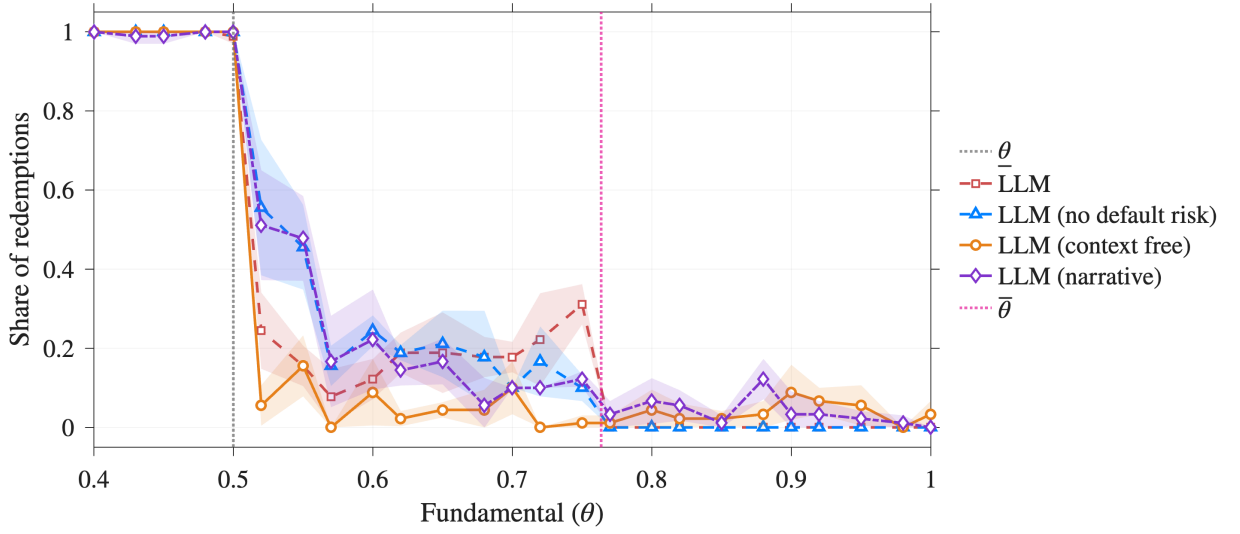


Figure C.1: Share of redemptions as a function of the fundamental θ . The line labelled “LLM” refers to the baseline case; “LLM (no default risk)” represents the case using Prompt C.1; “LLM (context free)” uses Prompt C.2; and “LLM (narrative)” uses Prompt C.3. Shaded areas show standard deviations constructed across three independent inference rounds. The vertical lines $\underline{\theta} = 0.5$ and $\bar{\theta} = 0.78$ are the bounds defined in Proposition 1.

Appendix D. Global games theoretical benchmark

In this appendix, we provide the details of the derivation of the threshold θ^* that characterizes the redemption strategy of investors receiving private signals. For the sake of clarity, remember that each investor i receives a private signal $\theta_i = \theta + \eta_i$, where η_i are i.i.d. noise drawn from $[-\eta, +\eta]$ and the parameter η represents a measure of information dispersion.

The proof adapts the standard approach in the global game literature (Goldstein and Pauzner, 2005; Chen et al., 2010) to the case with a discrete number of players. The arguments in their proof establish that there is a unique equilibrium in which investors redeem at $t = 1$ if and only if the signal they receive is below a common threshold θ^* , which is the signal at which an investor is indifferent between redeeming at $t = 1$ and $t = 2$ given what he or she believes about the signals received by other investors and, in turn, their behaviors.

We start by characterizing investors' decisions in two extreme ranges of fundamentals where investors have a dominant action. These two ranges correspond to the one characterized in Proposition 1. When $\theta < \underline{\theta}$ redeeming at $t = 1$ is a dominant action. When $\theta > \bar{\theta}$, redeeming at $t = 2$ is the dominant action.

Consider now the intermediate region where $\theta \in [\underline{\theta}, \bar{\theta}]$. Assume that investors behave according to a threshold strategy: they redeem their shares if they receive a signal below θ^* and stay until $t = 2$ otherwise.¹⁵ Given that the signal is uniformly distributed over the interval $[-\eta, +\eta]$, the probability of receiving a signal below θ^* is $\frac{\theta^* - \theta + \eta}{2\eta}$. Building on this, we can compute the share of investors receiving the signal below the cutoff, which is given by

$$w = \sum_{i=1}^A \{\theta_i < \theta^*\} \sim \text{Bin} \left(A, \frac{\theta^* - \theta + \eta}{2\eta} \right). \quad (\text{D.1})$$

It follows that the probability that W out of A investors have received a signal below θ^* is given by:

$$f(W, A) = \binom{A}{W} \left(\frac{\theta^* - \theta + \eta}{2\eta} \right)^W \left(1 - \frac{\theta^* - \theta + \eta}{2\eta} \right)^{A-W}. \quad (\text{D.2})$$

Using the above, we can compute the probability that the investor receiving the signal θ^* assigns to n out of $A - 1$ investors redeeming at $t = 1$ as:

$$\frac{\binom{A-1}{W}}{2\eta} \int_{\theta^* - \eta}^{\theta^* + \eta} \frac{(\theta^* - \theta + \eta)^W (\theta - \theta^* + \eta)^{A-1-W}}{(2\eta)^{A-1}} d\theta. \quad (\text{D.3})$$

This probability is equal to $\frac{1}{A}$ for every $W = 0, \dots, A-1$, a discrete counterpart of the Laplacian property in Morris and Shin (2002). Hence, the indifference condition that characterizes the run threshold θ^* reads:

$$\sum_{W=0}^{A-1} \frac{1}{A} \frac{N - W(1 + \lambda)}{N - W} R_1 R \left(\theta^* + \eta \frac{A - 1 - 2W}{A + 1} \right) = R_1, \quad (\text{D.4})$$

¹⁵This comes at no loss of generality, as Goldstein and Pauzner (2005) show that in this game every equilibrium strategy is a threshold strategy.

which gives

$$\theta^* \equiv \frac{A}{\sum_{W=0}^{A-1} \frac{N-W(1+\lambda)}{N-W} R} - \frac{\eta}{A+1} \frac{\sum_{W=0}^{A-1} (A-1-2W) \frac{N-W(1+\lambda)}{N-W}}{\sum_{W=0}^{A-1} \frac{N-W(1+\lambda)}{N-W}}. \quad (\text{D.5})$$

Finally, the average share of investors redeeming is:

$$\mathbb{E}[W] = \sum_{w=0}^A w \binom{A}{w} p^w (1-p)^{A-w}, \quad (\text{D.6})$$

where $p = \frac{\theta^* - \theta + \eta}{2\eta}$. Using the identity $w \binom{A}{w} = A \binom{A-1}{w-1}$, we obtain:

$$\mathbb{E}[W] = \sum_{w=1}^A A \binom{A-1}{w-1} p^w (1-p)^{A-w} = Ap \sum_{w=1}^A \binom{A-1}{w-1} p^{w-1} (1-p)^{(A-1)-(w-1)} = Ap, \quad (\text{D.7})$$

where the second-to-last step comes from using the binomial theorem. Then, we obtain

$$w^*(\theta, \theta^*) = \frac{\mathbb{E}[W]}{A} = \begin{cases} 0 & \text{if } \theta > \theta^* + \eta, \\ \frac{\theta^* - \theta + \eta}{2\eta} & \text{if } \theta \in [\theta^* - \eta, \theta^* + \eta], \\ 1 & \text{if } \theta < \theta^* - \eta. \end{cases} \quad (\text{D.8})$$